

**LLM-DRIVEN EXTRACTION AND FINANCIAL APPLICATION OF
MERGERS AND ACQUISITIONS EVENTS AND ACTORS FROM
EDGAR**

by
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**LLM-DRIVEN EXTRACTION AND FINANCIAL APPLICATION OF
MERGERS AND ACQUISITIONS EVENTS AND ACTORS FROM
EDGAR**

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ABSTRACT

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This thesis studies the private phase of mergers and acquisitions by introducing a novel large language model-based framework to extract key negotiation events and actors from SEC disclosures and analyzes how these private-phase dynamics shape competition, pricing, and renegotiation outcomes. It addresses a gap in the literature by replacing traditional hand-collected, static measures with scalable actor- and event-level data constructed from narrative filings with LLMs. The evidence shows that bidder participation and external advisor involvement have increased over time, accompanied by more price-specific offers and price revisions. The magnitude and frequency of price revisions are systematically related to the number of bidders, indicating that competition influences price formation. Negotiations follow a structured and economically intuitive sequence rather than unfolding as a random process. The thesis contributes methodologically by demonstrating how narrative corporate disclosures can be transformed into economically meaningful structured data using LLMs and substantively by providing large-sample evidence that opens the “black box” of private-phase M&A negotiations.

ÖZET

EDGAR BELGELERİNDEN BÜYÜK DİL MODELLERİ İLE BİRLEŞME VE SATIN ALMA OLAYLARI İLE AKTÖRLERİNİN ÇIKARILMASI VE FINANSAL UYGULAMALARI

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Bu tez, SEC bildirimlerinden birleşme ve satın alma işlemlerine ilişkin temel müzakere olaylarını ve aktörlerini çıkarmak için yeni bir büyük dil modeli tabanlı gerçeğe geliştirerek birleşme ve satın almaların özel aşamasını incelemekte ve bu özel aşama dinamiklerinin rekabeti, fiyatlamayı ve yeniden müzakere sonuçlarını nasıl şekillendirdiğini analiz etmektedir. Çalışma, geleneksel elle toplanmış ve statik ölçütlerin yerine, anlatı niteliğindeki dosyalardan büyük dil modelleri kullanılarak oluşturulan ölçeklenebilir aktör ve olay düzeyindeki verileri koyarak literatürdeki bir boşluğu doldurmaktadır. Bulgular, teklif verenlerin katılımının ve dış danışmanların sürece dahil olmasının zaman içinde arttığını; buna daha fazla fiyata özgü teklif ve fiyat revizyonunun eşlik ettiğini göstermektedir. Fiyat revizyonlarının büyüklüğü ve sıklığı, teklif veren sayısı ile sistematik biçimde ilişkilidir; bu da rekabetin fiyat oluşumunu etkilediğine işaret etmektedir. Müzakereler, rastgele bir süreç gibi ilerlemekten ziyade, yapılandırılmış ve ekonomik açıdan sezgisel bir sıralama izlemektedir. Tez, anlatı niteliğindeki kurumsal açıklamaların büyük dil modelleri kullanılarak ekonomik açıdan anlamlı yapılandırılmış verilere dönüştürülebileceğini göstererek metodolojik katkı sağlamakta; ayrıca özel aşama birleşme ve satın alma müzakerelerinin “kara kutusunu” açan geniş örneklemli kanıtlar sunarak içerik açısından da katkıda bulunmaktadır.

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1. INTRODUCTION

The private phase of mergers and acquisitions (M&A), documented primarily in the background sections of SEC filings, has become an important source of information on negotiation processes in corporate transactions. Early work by A. L. Boone and Mulherin, 2007 demonstrated that competition in the M&A market is more intense than previously assumed in the public phase and highlighted the value of the background section for studying bidder interactions, formal offers, and confidentiality agreements. Subsequent studies have constructed hand-collected datasets to examine the private phase, including A. L. Boone and Mulherin, 2008, Aktas et al., 2009, Bowen et al., 2018, Liu and Mulherin, 2019, Liu et al., 2022, and A. Boone et al., 2024. While these studies provide valuable insights, their reliance on manually curated samples limits scale and time coverage, as those studies filtered deals and sampled deals from specific date ranges.

The literature on private-phase M&A negotiations faces two key fundamental gaps. First, existing datasets are manually constructed, which constrains sample size and introduces potential selection and human biases. Second, although AI-based methodologies are increasingly used in M&A research, the private phase remains largely unexplored due to the unstructured and lengthy nature of these filings. A comprehensive, long-horizon analysis of private-phase negotiations using large language models (LLMs) and advanced NLP can address these limitations. This thesis bridges these gaps by applying LLM-based extraction pipelines to a long-horizon sample of S-4, PREM14A, DEFM14A, and tender-offer filings from 1994 to the end of the first quarter of 2025 to extract structured data at scale and conduct downstream analyses for financial applications.

Specifically, we extract introductory parts and background-of-the-transaction narratives from SEC filings and use an LLM to (a) identify key participants in the private phase (target company, final acquirer company, competing bidders, and external financial and legal advisors); (b) detect and classify important negotiation events; and

(c) extract dollar-denominated price mentions using the spaCy `en_core_web_trf` transformer model¹ and price revisions from price-specific offer events. This approach enables a systematic reconstruction of negotiation sequences, participant structures, and price dynamics across thousands of transactions.

The contribution of this study is to bridge the intersectional gap in the literature by covering a longer time span for analyzing the private phase of mergers and acquisitions using a multistage LLM-based framework applied to lengthy narrative SEC disclosures. This thesis addresses four research questions:

1. How do bidder competition and advisor involvement shape negotiation dynamics?
2. How do negotiation events unfold during the private phase of mergers and acquisitions?
3. How do private-phase negotiation events translate into offer price formation and renegotiation outcomes?
4. Can large language models reliably extract economically meaningful negotiation data from SEC narrative disclosures?

The main findings of this study are fourfold:

First, the intensity of private-phase activity in mergers and acquisitions has increased substantially over time. The number of competing bidders and external financial and legal advisors per transaction has risen steadily over the sample period, and procedural negotiation events, particularly price-specific offers and price bids, have become more frequent. This evidence suggests that the private phase of M&A has become more active and competitive.

Second, private-phase negotiations follow a structured and economically intuitive sequence. Event-time distributions and transition matrices reveal a consistent ordering in which targets first retain financial advisors, followed by the execution of confidentiality agreements, the submission of price-specific offers by bidders, and ultimately the public announcement of the transaction (*Target Financial Advisor Hiring* → *Confidentiality Agreement* → *Price-Specific Offer by Bidders* → *Deal Announcement*). This ordering indicates that negotiation processes in M&A are not random but instead unfold through

1. https://huggingface.co/spacy/en_core_web_trf

procedural stages that reflect contracting frictions, information acquisition, and sequential bargaining constraints.

Third, private-phase events are associated with economically meaningful variation in formal price bids and renegotiation dynamics. Bidder competition is associated with more frequent price revisions; for per-share prices, the presence of at least one competing bidder coincides with larger upward revisions and more pronounced price jumps. However, conditional on revision activity, the mean and median per-share price revisions decline with the number of bidders, indicating that bidders respond to intensified competition by adjusting prices in smaller, more conservative increments. This pattern is consistent with bidders internalizing winner’s curse concerns and strategically submitting more conservative revisions to lower the extra premium paid. Price-specific offers exhibit a clear bimodal distribution, consistent with the coexistence of per-share prices and total transaction values. The increase in the number of disclosed prices and revision counts per transaction over time indicates rising disclosure intensity and more active renegotiation processes. Initial offer prices serve as reference points for subsequent pricing, as cumulative revisions relative to the first offer display non-monotonic and asymmetric patterns. In contrast, the magnitude and direction of consecutive price revisions are smaller and more symmetric, respectively.

Fourth, large language models can reliably extract structured negotiation data from narrative SEC disclosures at scale. The extracted event sequences and participant identities exhibit economically sensible patterns and closely match structured SDC records, although EDGAR filings systematically underreport advisor involvement relative to SDC. Validation is further strengthened by human annotation. Overall, annotation demonstrates that LLM-based text extraction yields high-quality negotiation data suitable for large-sample empirical research, enabling the study of private-phase M&A dynamics beyond small hand-collected samples.

Taken together, these results provide the first long-horizon, large-scale evidence on the structure, timing, and pricing consequences of private-phase M&A negotiations. By combining LLMs with SEC narrative disclosures, this study opens new empirical insights into negotiation processes that have traditionally been unobservable and establishes a scalable framework for future research on the private phase of mergers and acquisitions.

The remainder of the thesis documents data sources, extraction methods, validation, and empirical tests. Chapter 2 discusses the relevant literature. It focuses on both papers related to the private phase of mergers and acquisitions and on papers using

AI for mergers and acquisitions in general. chapter 3 illustrates which data were used, how they were downloaded, data filtering, specific parts and sections extracted for downstream tasks, and a summary of the final dataset used. chapter 4 elaborates on the methodology used for extracting valuable information from SEC filings. It explains how an LLM is used for extracting parties and events and categorizing the events, how price revision analysis was conducted on *price-specific offers by bidders' events*, the annotation procedure, and the matching criteria and procedure between SDC and EDGAR datasets. chapter 5 discusses the empirical results done after information extraction steps. The focus was on the distribution and trends of parties, temporal and category analysis of events, price-based bids, and validation using matched EDGAR–SDC deal data.

2. RELATED WORKS

In this section, related works are categorized into three main sections. The first section analyzes the motivations behind mergers and acquisitions. The second section emphasizes the *private phase of mergers and acquisitions* through the lens of deal initiation, target and acquirer matching, the role of advisors, private negotiation and bidding, and factors affecting target company premiums. The third section provides papers that used artificial intelligence and data mining techniques as their methodologies.

2.1 Motives for Mergers and Acquisitions

Mergers and acquisitions are motivated by synergy, market power, behavioral, and agency motives Aktas and Boone, 2023. Healy et al., 1990 found that merged firms experience improvements in operating cash flow and abnormal stock returns while preserving their capital expenditures and R&D rates. Studies of bank mergers have shown that value gains are related to cost reduction from eliminating overlapping operations Houston et al., 2001. Devos et al., 2009 reported that synergies in completed deals arise from deployment efficiencies of economic resources, with operational synergies on average exceeding financial synergies. Maksimovic et al., 2011 documented that target restructuring by acquirers tends to improve resource allocation: companies redraw their boundaries by selling or maintaining acquired plants to increase productivity. Productive firms tend to become public earlier, and public companies undertake acquisitions with greater diligence than private firms. Assets acquired by public companies are more likely to experience productivity gains, especially when acquired from other public firms Maksimovic et al., 2010. Gorbenko and Malenko, 2014 showed that various types of target firms attract both strategic and financial

bidders, challenging the notion that strategic buyers value targets only for synergies.

Behavioral motives are also important: Malmendier and Tate, 2008 demonstrated that overconfident CEOs are more active in acquisitions and are more likely to undertake more value-destroying deals. Rhodes-Kropf and Viswanathan, 2004 argued that misvaluation can generate merger waves: overvaluation increases the probability of mergers, while undervaluation can halt waves. Social motives and imitation were emphasized by M. Yang and Hyland, 2006, who categorized imitation in M&As at the firm, market, and industry levels. Agency-driven behavior and managerial herding were shown to be particularly evident during merger waves Duchin and Schmidt, 2013, and defensive managerial motives were discussed by Gorton et al., 2005: managers become more acquisitive to reduce the chance of being a target for other firms.

2.2 Private Deal Formation and Negotiation in M&As

The private phase of mergers and acquisitions begins when firms reach out to each other to show interest; this stage can be initiated by either the target or the acquirer company. External parties such as investment banks can also initiate such processes on behalf of companies Aktas and Boone, 2023.

2.2.1 Determinants of Deal Initiation

Aktas et al., 2009 documented that higher institutional ownership and a higher target Tobin's Q reduce the probability that the target firm initiates its sale. Masulis and Simsir, 2018 found that, between 1997 and 2012, approximately thirty-five percent of deals were initiated by target companies; target initiation was mainly associated with financial and economic distress, binding financial constraints, and macroeconomic recessions. Fidrmuc and Xia, 2019 contributed to the understanding of CEO ownership and deal initiation, concluding that a higher rate of CEO ownership motivates managers to initiate the sale of their company and offer higher golden parachutes, stock, and stock option grants.

2.2.2 Acquirer–Target Pairing and Synergy Formation

Jovanovic and Rousseau, 2002 extended the Q-theory of investment to M&A activities, showing that firms’ merger and acquisition investment is sensitive to Tobin’s Q. Maksimovic and Phillips, 2001 found that a firm’s probability of buying additional assets increases with its efficiency and size; firms are more likely to sell relatively unproductive divisions when the selling firm itself is less productive. Harford, 1999 showed that cash-rich firms are engaged more in acquisition activity than other firms. A firm’s debt rating also influences a company’s ability to acquire and its acquisition quality; rated firms are more able to acquire and tend to complete higher-quality acquisitions than nonrated firms.

2.2.3 The Role of Financial and Legal Advisors

Investment banks and law firms are central external actors in the takeover process. Investment banks assist companies in finding investment opportunities and with valuation, financing, and managing the sale process. Legal advisors support fiduciary duties, legal due diligence, and legal documents and agreement drafting Aktas and Boone, 2023. Servaes and Zenner, 1996 showed that the use of an investment bank depends on deal complexity, transaction type, the acquirer’s prior acquisition experience, and the target’s degree of diversification. Bao and Edmans, 2010 found a positive association between investment bank involvement and announcement returns for acquisitions. Their results demonstrated that certain banks can identify bad deals, contrasting previous findings that bank quality does not matter in M&A outcomes. Golubov et al., 2012 demonstrated that top-tier financial advisors relate to higher bidder gains in public acquisitions but not in subsidiary or private transactions. Their advantage stems from identifying higher-synergy mergers (consistent with the better merger hypothesis) and helping bidders capture more of those synergies. Krishnan and Masulis, 2013 reported that top-tier bidder law firms are linked to higher transaction completion rates and larger takeover premia than less prominent legal advisors. On the other hand, top-tier target law firms tend to have lower completion rates but secure higher premia. The authors concluded that top law firms have a stronger influence on M&A outcomes than top investment banks, suggesting potential omitted-variable bias in earlier studies that overlooked law-firm effects.

2.2.4 Private Negotiation and Bidding Dynamics

A. L. Boone and Mulherin, 2007 were among the first to focus explicitly on the private phase of mergers and acquisitions using the background sections of SEC merger documents (Proxy 14A, S-4, 14D) from EDGAR, LexisNexis, and Laser Disclosure. Using 400 takeovers of large U.S. publicly traded firms announced between 1989 and 1999, they extracted information on potential buyers contacted, confidentiality/standstill agreements signed, private written bids, and public bids. They classify deals as auctions when multiple buyers contacted target firms and multiple confidentiality agreements were signed; deals with only a single private bidder were classified as negotiations. The authors reported that "public takeover activity is only the tip of the iceberg of actual takeover competition during the 1990s." In their sample, half of the companies experienced auction processes, and the rest were negotiated with a single company. In auctions, an average of 20.67 potential buyers were contacted, 6.83 confidentiality agreements were signed, and roughly 1.57 private bids and 1.24 bidders were observed.

The same authors A. L. Boone and Mulherin, 2008 tested the winner's curse hypothesis, which predicts that bidder returns decline as takeover competition intensifies and target valuation uncertainty rises; using hand-collected data on 308 U.S. takeovers announced between 1989 and 1999, they found no significant relation between bidder returns and competition level. Aktas et al., 2009, drawing on hand-collected data from 1,774 M&A transactions between 1994 and 2007 with available SEC filings (14-A, S-4, and 14-D), showed that latent competition increases bid premiums. Liu and Officer, 2019 referred to the private phase of mergers and acquisitions as a "black box," documenting that bid revision is common and that price revisions are associated with public market values of target firms.

Liu and Mulherin, 2019 analyzed 388 completed and withdrawn deals between 1981 and 2014 and found that competition for targets has remained roughly constant when both private and public auctions are considered, despite a decline in hostile bids and publicly reported takeover bidding. Liu et al., 2022 constructed a manually collected dataset of 841 deals announced from 2005 to 2019 and showed that many deals that start as auctions switch to negotiations with a single buyer at a later stage; they claimed that M&As are fluid, a phenomenon they term "negotiauction." A. Boone et al., 2024 argued that stronger initial bids streamline the private phase of mergers and acquisitions, resulting in fewer additional bidding rounds, smaller bid revisions, and a higher probability that the initial bidder completes the transaction. They also showed that stronger offers are associated with higher acquirer returns

when the deal is initiated by the target and with higher combined returns when the bidder initiates the transaction.

2.2.5 Target Premium Factors

The literature on target premiums suggests that takeover pricing is a culmination of multiple factors, including target fundamentals, information asymmetry, and transaction characteristics. Alexandridis et al., 2012 challenged the conventional view that larger targets require higher premiums by showing that target size is negatively related to offer premiums. Fidrmuc and Moeller, 2014 found that takeovers with more bidders exhibit lower premiums in the raw data but showed that this negative relationship is driven by underlying deal characteristics and disappears after controlling for differences across transactions.

Subsequent research emphasized the importance of advisory expertise; Agrawal et al., 2023 showed that the target firm’s financial advisors improve target valuation by reducing information disadvantages. Moreover, Chang et al., 2016 demonstrated that acquirers can exploit informational advantages by hiring target firms’ ex-advisors, resulting in lower premiums.

Furthermore, Lastunen, 2024 found that target profitability (ROA) is negatively associated with takeover premiums, and Liu et al., 2023 documented that tender offers are associated with higher premiums compared with negotiated transactions.

2.3 Artificial Intelligence in M&A: Data, Methods, and Deal Phases

The literature on artificial intelligence (AI) applications to mergers and acquisitions (M&A) has expanded rapidly over the past decade. Early contributions relied heavily on structured financial and deal-level variables and classical statistical or machine learning techniques (e.g., Lasso, logistic regression, random forests) to predict the likelihood or status of transactions Aaldering et al., 2019; Yan et al., 2016; B. Yang et al., 2019. From 2019, textual and alternative unstructured sources such as 10-K filings, conference calls, news and social media feeds, and patent texts were

utilized. More complex methods such as regression, naive Bayes, LSTM, PCA, kNN, neural networks, TF-IDF, and dictionary methods were used. Bi and Zhang, 2021; Jiang, 2021; Karatas and Hirs, 2021b; Schiller, 2021; Venuti, 2021. More recent work (2022–2025) has demonstrated a clear tendency toward graph-based and transformer-based approaches and LLMs. By using graph-based neural networks, the aim was to predict target companies and model relationships among firms, and by using LLMs, a new paradigm has emerged. Bozman et al., 2025; Cheng et al., 2025; Degen et al., 2024b; Krause, 2023; L. Li, 2025; Lohmeier et al., 2023; D. Yang, 2024

Across methodological choices, the work can be grouped into several broad objectives: (i) deal prediction and target recommendation; (ii) post-merger outcome and performance forecasting; and (iii) automated due diligence and document understanding. Data sources have reflected these objectives, including structured financial databases (Compustat, CRSP, SDC), alternative provider feeds (Refinitiv/RavenPack/MarketPsych); SEC filings and earnings calls (EDGAR/Capital IQ); and firm-level graph data (CrunchBase, Wikidata). Data type proliferation has driven methodological diversity, ranging from classical ML and econometric models to temporal networks, statistical models, deep learning-based models, and transformer-based LLMs.

Table 2.1 documents M&A papers with their title, year of publication, M&A phase, data used, and methodologies used in chronological order. This approach shows the shift towards deep learning models and LLMs and from tabular to textual data.

Table 2.1 Overview of Data Mining and AI Methods Applied Across M&A Phases

Paper Title	Year	M&A Phase	Data Used	Method Used
Leveraging Artificial Intelligence for Advanced Deal Sourcing in U.S. Mergers and Acquisitions to Improve Financial Efficiency Asamoah and Laryea, 2025	2025	Pre-deal	Not specified	AI-driven conceptual framework
Enhancing the Merger Simulation Toolkit with ML/AI Chiang et al., 2025	2025	Pre-deal	Market-level prices, quantities, product characteristics, and cost shifters (airline DB1B database)	Variational Method of Moments (VMM) with neural networks

Continued on next page

Paper Title	Year	M&A Phase	Data Used	Method Used
Predicting mergers & acquisitions: A machine learning-based approach Zhao et al., 2025	2025	Pre-deal	Structured financial, governance, managerial, industry, and macroeconomic variables (China dataset)	Decision trees; random forests; SVM; Naive Bayes; KNN
Better Than a Coin Flip? Merger Success and Artificial Intelligence Models Bozman et al., 2025	2025	Post-deal	Deal-level and firm-level structured variables from SDC, Compustat, and CRSP	Ridge; Lasso; Random forest; Gradient Boosting; GPT-4
Fine-tuning Transformer Models for M&A Target Prediction in the U.S. Energy Sector Rodríguez-Muñoz-de-Baena et al., 2025	2025	Pre-deal	Unstructured textual data from corporate annual reports	Fine-tuning BERT-family transformer models
Decision-Making in M&A Under Market Mispricing: The Role of Deep Learning Models Tang, 2025	2025	Pre-deal	Sentiment survey data; synthetic financial dataset with technical indicators and macroeconomic factors	Deep learning models (CNN, LSTM)
Construction of knowledge graph for enterprise mergers and acquisitions: cross-domain value mining method of large language model (LLM) and graph neural network L. Li, 2025	2025	Pre-deal	Multimodal heterogeneous data (text, numeric, structural) for enterprise knowledge graph construction	Integrated LLM-GNN framework (E-GraphNet; DAPR)
A Knowledge Graph-Based Target Recommendation Approach for Mergers and Acquisitions Cheng et al., 2025	2025	Pre-deal	Firm attribute graph constructed from CrunchBase	KG embeddings (TransE/TransR/TransD/CrossE); RED-GNN
Computational Simulation of Hypermarket Merger: Statistical Analysis, AI and Big Data in EV Gabriel do Breviario et al., 2025	2025	Post-merger	Financial statements (income statements, balance sheets), stock market data (e.g., Investing.com), firm-level financial variables (WACC, growth rate, reinvestment rate)	Monte Carlo simulation, multiple regression, and neural networks
Continued on next page				

Paper Title	Year	M&A Phase	Data Used	Method Used
Optimizing Target Identification in the U.S. Capital Market Mergers and Acquisitions through Artificial Intelligence: Implications for Financial Efficiency, Compliance, and National Economic Competitiveness Akingbade and Agbeve, n.d.	2025	Pre-deal	Structured data (financial statements, SEC filings) and unstructured data (news, contracts, legal and regulatory documents)	Machine learning, natural language processing (NLP), predictive analytics, automated due diligence systems
The Role of Artificial Intelligence in Enhancing Decision-making and Efficiency in Mergers and Acquisitions: A Case Study Approach within the U.S. Capital Market Narteh-Kofi et al., n.d.	2025	Pre-deal; deal execution; post-merger integration	Structured financial records and unstructured data (legal documents, news articles, market and industry reports)	Machine learning, predictive analytics, natural language processing (NLP), AI-enabled automation and decision-support systems
Artificial Intelligence on Merger and Acquisition Processes: Observation from the Target Identification and Due Diligence Perspective Rahman, 2025	2025	Pre-deal	Structured firm and deal data and unstructured documents used for target screening and due diligence	Machine learning, natural language processing (NLP), predictive analytics
AI-Driven M&A Forecasting: Singular Spectrum Analysis Model in the EDA Industry Zylfiu and Marinova, 2025	2025	Pre-deal (industry consolidation forecasting and strategic timing)	Historical M&A transaction data in the EDA industry; proprietary TechImpactScore; aggregated time-series data	Singular Spectrum Analysis (SSA) time-series decomposition and forecasting
Machine learning in bank merger prediction: A text-based approach Katsafados et al., 2024	2024	Pre-deal	Bank annual reports (financial and textual disclosures)	SVM; logistic regression; random forest; MLP; TF-IDF
What is the Deal?: Predicting M&A Outcomes with Machine Learning Campbell et al., 2024	2024	Post-merger	Financial fundamentals and deal characteristics from CRSP, Compustat, and SDC	Random Forest; Gradient Boosting; Lasso and Neural Networks
Pricing media sentiment: Evidence from global mergers and acquisitions Shams et al., 2024	2024	Pre-deal and Deal Process	Financial news sentiment and news-flow intensity from RavenPack (5,196 global deals, 2000–2018)	RavenPack sentiment; news intensity models
Predicting M&A Targets Using News Sentiment and Topic Detection Hajek and Henriques, 2024	2024	Pre-deal	Financial news articles with sentiment and extracted topics	Sentiment polarity; topic detection features

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Paper Title	Year	M&A Phase	Data Used	Method Used
Leveraging Natural Language Processing and Large Language Models for Assisting Due Diligence in the Legal Domain Jang and Stikkel, 2024	2024	Pre-deal (legal)	Legal document corpus from the KIRA dataset	BERT; LegalBERT; Bi-LSTM; GPT-4
AI-Driven M&A Target Selection and Synergy Prediction: A Machine Learning-Based Approach Zhang et al., 2024	2024	Pre- & post-merger	Structured financial/deal variables and unstructured company descriptions/press releases (10,000 deals)	A hybrid ensemble model of gradient boosting, SVMs, and neural networks; TF-IDF
A Deep Learning Method for Predicting M&As: Temporal Dynamic Industry Networks D. Yang, 2024	2024	Pre-deal	Temporal industry-network dataset	Dynamic temporal network model integrated with deep learning techniques
Large language models and M&A: Can ChatGPT help forecast M&A activity? Degen et al., 2024a	2024	Pre-deal	Earnings conference call transcripts	GPT-4-based sentiment scoring (MASS)
AI in Mergers and Acquisitions: Analyzing the Effectiveness of Artificial Intelligence in Due Diligence Bedekar et al., 2024	2024	Deal Process	Structured financial data, unstructured alternative data (e.g., website traffic, hiring trends, social sentiment, patent filings), and proprietary databases of historical transactions	Machine learning (ML), natural language processing (NLP), and predictive modeling
Predict Industry Merger Waves Utilizing Supply Network Information Qu et al., 2024	2024	Pre-deal	Firm-level financial statements and supply network data (triadic relationships among firms)	Random Forest, support vector machines (SVM), network-based feature engineering
Determinants of Successful Mergers and Acquisitions in China: Evidence from Machine Learning Zhou et al., 2024	2024	Pre-deal; deal execution	Deal characteristics, firm-level financial metrics, non-financial managerial indicators, external policy uncertainty variables (China)	Gradient Boosting (GBM), Random Forest (RFM), SVM, KNN, SMOTE/ADASYN
<i>Continued on next page</i>				

Paper Title	Year	M&A Phase	Data Used	Method Used
Predicting Cross-border Merger and Acquisition Completion through CEO Characteristics: A Machine Learning Approach Cheng and Dai, 2025	2024	Pre-deal	CEO demographic and professional background data (BoardEx), CEO personality traits extracted from earnings call transcripts, deal-level information	LightGBM, SHAP-based model interpretation, textual analysis for personality inference
AI in Mergers and Acquisitions: Analyzing the Effectiveness of Artificial Intelligence in Due Diligence Bedekar et al., 2024	2024	Pre-deal; deal execution (due diligence)	Structured financial data, unstructured legal documents, news articles, social media data, market data	Machine learning (Random Forest, SVM), NLP (BERT), predictive analytics (K-means, logistic regression)
Identifying M&A Targets from Textual Disclosures: A Transformer Neural Network Approach Lohmeier et al., 2023	2023	Pre-deal	10-K business descriptions and Management Discussion & Analysis sections from SEC EDGAR	RoBERTa; Longformer
Optimizing M&A Outcomes: Harnessing the Power of Big Data Analytics and Natural Language Processing Parikh et al., 2023	2023	Deal execution (due diligence); post-merger integration	Unstructured textual data from legal documents, financial statements, internal and external communications	Natural language processing (NLP), transformer-based models (BERT, GPT-3), named entity recognition, text summarization, machine learning
Leveraging Big Data for M&A: Towards Designing Process Mining Analyses for Process Assessment in IT Due Diligence Analyses for Process Assessment in IT Due Diligence Eggers et al., 2023	2023	Pre-deal	ERP event logs (SAP; O2C and P2P processes)	Process mining (PM), a particular type of data mining technique, is applied to event log data sourced from ERP systems
ChatGPT and Other AI Models as a Due Diligence Tool: Benefits and Limitations for Private Firm Investment Analysis Krause, 2023	2023	Pre-deal	Unstructured textual data from SEC filings, news, and social media	General-purpose LLMs; NLP and sentiment analysis
Combination of Artificial Intelligence with Mergers and Acquisitions Wang and Zhou, 2023	2023	Pre-deal; deal execution; post-merger integration	Financial documents, legal and contractual texts, internal M&A documentation, other unstructured textual data	Natural language processing (NLP), machine learning, AI-enabled document analysis and risk assessment tools

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Paper Title	Year	M&A Phase	Data Used	Method Used
The Application of Feed-Forward Neural Networks to Merger Arbitrage: A Return-Based Analysis Braun et al., 2023	2023	Pre-deal	Deal-level characteristics, target firm attributes, and macroeconomic indicators	Feed-forward neural networks (FFNN), factor selection for return optimization
Combination of Artificial Intelligence with Mergers and Acquisitions Wang and Zhou, 2023	2023	Pre-deal; deal execution; post-merger integration	Financial documents, legal and contractual texts, internal M&A documentation, other unstructured textual data	Natural language processing (NLP), machine learning, AI-enabled document analysis and risk assessment tools
Application of Machine Learning Models for Predictions on Cross-Border Merger and Acquisition Decisions with ESG Characteristics from an Ecosystem and Sustainable Development Perspective Hong et al., 2022	2022	Pre-deal	Unstructured disclosures (SEC filings, news, social media) with ESG-related factors	Machine learning and NLP (sentiment analysis)
Graph Neural Network for Merger and Acquisition Prediction Y. Li et al., 2022	2022	Pre-deal	Structured entity data from CrunchBase (companies, persons, investors) and unstructured text data from TechCrunch articles	Spectral/spatial GNNs; XGBoost, Logistic Regression, and SVM
Application of Machine Learning Models for Predictions on Cross-Border Merger and Acquisition Decisions with ESG Characteristics from an Ecosystem and Sustainable Development Perspective Hong et al., n.d.	2022	Pre-deal	Large-scale M&A dataset (215,160 deals), firm-level financial statements, ESG scores, country-, deal-, and firm-level indicators	AdaBoost (decision trees), support vector machines (SVM), logistic regression
Mergers and Acquisitions Prediction Using Hybrid Machine Learning and Deep Learning Approach Ghadekar et al., 2022	2022	Pre-deal	Structured firm-level data from CrunchBase (funding rounds, founder education, IPO status)	Hybrid ML framework (Gradient Boosting, SVM, Multi-Layer Perceptron), RANSAC-based imputation
Post Mergers and Acquisitions Integration Solution Using Machine Learning Maan and Nagwekar, 2022	2022	Post-merger integration (PMI)	Unstructured questionnaire responses across finance, IT, and HR functions	TF-IDF vectorization, cosine similarity (unsupervised learning), gap and synergy analysis

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Paper Title	Year	M&A Phase	Data Used	Method Used
Prediction and Visualization of Mergers and Acquisitions Using Economic Complexity Arsini et al., 2023	2022	Pre-deal	Patent data, industrial sector classifications, technology proximity and co-occurrence networks	Cosine similarity (Jaffe metric), network-based co-occurrence analysis, Random Forest, economic complexity measures
Can Digital Transformation Facilitate Firms' M&A: Empirical Discovery Based on Machine Learning Tu and He, 2023	2022	Pre-deal	Unstructured textual data from annual reports (MD&A sections), firm-level characteristics	Word2Vec-based text embedding, machine learning-based index construction, empirical causal analysis
Using Machine Learning to Analyze Merger Activity Jiang, 2021	2021	Pre-deal	10-K filings from SEC EDGAR	L2 Lasso; TF-IDF, and bag-of-words
Are Managers listening to Twitter? Evidence from Mergers & Acquisitions Schiller, 2021	2021	Pre-deal and Deal Process	Twitter posts (firm- and deal-related sentiment time series)	Naive Bayes; time-series analysis
Predicting Status of Pre and Post M&A Deals Using Machine Learning and Deep Learning Techniques Karatas and Hirs, 2021b	2021	Pre- & post-deal	Structured financial and deal variables from FactSet; sentiment time series from Refinitiv MarketPsych	Feedforward neural networks (FFNN); LSTM autoencoders; PCA, MCA, kNN
Predicting Mergers and Acquisitions using Graph-based Deep Learning Venuti, 2021	2021	Pre-deal	Firm knowledge graph built from Wikidata triples (over 61k companies and 115k edges)	GraphSAGE framework
Investigating Insensitivity to Prior Probabilities in Merger and Acquisition (M&A) Decision Making Mcgaughan and Chengalur-Smith, 2021	2021	All stages (decision-making across M&A evaluation)	Global panel data on M&A transactions; published failure-rate benchmarks (e.g., KPMG 1999 study)	Regression Discontinuity in Time; Two-way Fixed Effects models
Forecasting mergers and acquisitions failure based on partial-sigmoid neural network and feature selection Bi and Zhang, 2021	2021	Pre-deal	Structured financial and deal variables from iFinD and CSMAR	Partial-sigmoid neural network (PSNN) and feature selection methods (chi-square test, information gain, GBDT)
Predicting Status of Pre- and Post-M&A Deals Using Machine Learning and Deep Learning Techniques Karatas and Hirs, 2021a	2021	Pre-deal (post-announcement)	Financial ratios, deal characteristics (e.g., payment method, management attitude), sentiment scores from sequential data	kNN imputation, PCA/MCA, LSTM autoencoders, feedforward neural networks (FFNN), SMOTE

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Paper Title	Year	M&A Phase	Data Used	Method Used
What's Really in a Deal? Evidence from Textual Analysis of M&A Conference Calls Hu et al., 2020	2020	Post-deal	Earnings call transcripts from Capital IQ (814 calls, 2011–2018); deal data from SDC Platinum; firm data from CRSP and Compustat	Dictionary-based sentiment; regex extractions; probit/Heckman models
Recommending untapped M&A opportunities: A combined approach using principal component analysis and collaborative filtering Aaldering et al., 2019	2019	Pre-deal	M&A transactions from SDC Platinum (4,688 biotech deals, 2012–2016)	PCA + Collaborative Filtering
Can financial media sentiment predict merger and acquisition performance? B. Yang et al., 2019	2019	Pre-deal and Deal Process	Financial news articles (around 480,000 articles) covering 288 U.S. acquisitions	Dictionary-based sentiment; event-study regressions
Analyzing Mergers and Acquisitions (M&A) in Japan Using AI Methods Shao et al., 2018	2018	Pre-deal	Firm-level financial data (cash flow characteristics), industry classification, historical M&A relationship data (Japan)	Unsupervised learning (K-means clustering), Principal Component Analysis (PCA)
Modeling Contagious Merger and Acquisition via Point Processes with a Profile Regression Prior Yan et al., 2016	2016	Pre-deal	Startup and firm attribute data from CrunchBase	Multidimensional Hawkes Process; logistic regression; Latent Dirichlet Allocation (LDA)
Identification and Evaluation of Corporations for Merger and Acquisition Strategies Using Patent Information and Text Mining Park et al., 2013	2013	Pre-deal	Unstructured patent text data	Text mining; NLP (for SAO structure extraction); semantic similarity measurement; cluster analysis; multidimensional scaling (MDS)
Implications of data screens on merger and acquisition analysis: A large sample study of mergers and acquisitions from 1992-2009 Netter et al., 2011	2010	Pre-deal	Structured transactional data (SDC Platinum: deal value, firm status, payment method, industry codes)	Do not directly utilize data mining methods, yet critically analyze how data screening and sample selection influence M&A studies, which is fundamental for any data-centric or data mining technique

3. DATA ACQUISITION AND EXTRACTION METHODOLOGY

This chapter describes the data sources, data collection approach, and extraction methodologies employed to construct the textual datasets used in the subsequent LLM-based framework.

3.1 Data Sources

This study relies on two primary data sources to capture detailed information on mergers and acquisitions: EDGAR filings from the U.S. Securities and Exchange Commission and the SDC Platinum database.

3.1.1 EDGAR SEC Filings (U.S. Securities and Exchange Commission)

The Electronic Data Gathering, Analysis, and Retrieval (EDGAR) system is the SEC’s central platform for collecting and disseminating corporate filings submitted under U.S. federal securities laws. Public companies are required to file financial statements, annual and quarterly reports (e.g., 10-K, 10-Q), and merger- and acquisition-related documents such as 8-K, SC 14D-9, Form S-4, and proxy statements.

3.1.2 SDC Platinum (Refinitiv / Thomson One)

SDC Platinum is a widely used commercial database maintained by Refinitiv (formerly Thomson Reuters) that provides comprehensive coverage of global mergers and acquisitions. It includes standardized deal-level information such as announcement dates, transaction values, deal type, advisor information, and deal outcomes.

3.2 EDGAR Data Collection

To collect mergers and acquisitions EDGAR filings, the Edgar Archive Index ¹ was used to crawl all years and their corresponding quarters. After downloading all **form.idx** files for each year and quarter, those files were read and filtered by form type. We focused on **DEFM14A**, **PREM14A**, **SC 14D9**, **SC TO-T**, and **S-4** filings because these documents typically contain the "**background of the merger**" and "**background of the transaction**" sections of mergers and acquisitions, discussing the story behind those deals. For each year and quarter, Excel dataframes were constructed, focusing on the mentioned forms and constructing URLs for later steps. In the end, dataframes included **Form Type**, **Company Name**, **CIK**, **Date Filed**, and **File Path** columns used in subsequent steps. In the second step of data collection, these URLs were used to send HTTP requests. With the help of asynchronous programming and rate-limited requests, all related mergers and acquisitions files limited to TXT, HTM, and HTML formats were downloaded for each transaction and saved in organized subdirectories. In total, the downloaded dataset comprised approximately **1 terabyte** of data. Table 3.1 summarizes the number of files and their sizes by year.

3.3 File Filtering and Identification of Relevant Documents

After downloading the relevant Edgar filings into a structured directory hierarchy according to the previous step, files were filtered and chosen based on the following

1. <https://www.sec.gov/Archives/edgar/full-index/>

criteria: (i) a minimum length threshold of **1,000** words to ensure capturing the main files, not extensions to agreements; and (ii) the presence of the keyword **background**, which is typical of M&A disclosure sections describing deal history, bidder interactions, and negotiation processes. In this way, qualifying files were filtered and saved for subsequent analysis, text extraction, and LLM processing. For deals with more than one qualified document, multiple qualifying documents were saved and indexed sequentially (e.g., _1, _2, _3) for later stages. Table 3.2 summarizes the after-filtering data statistics, which in total amount to approximately 160.7 GB of data. For clarification, the transition from TXT files to HTML-formatted files was not abrupt: as more qualified files appeared after 2003, it became unnecessary to focus on TXT files beyond 2006.

3.4 Detection of Background Section Titles

When dealing with TXT and HTML files from Edgar filings, there was no specific and standard way of detecting the **background of the deal** section. To overcome this difficulty, focus was driven towards HTML files, as they encompassed more of the filings, especially in recent years, than TXT files. For dealing with HTML files, all `<a>` tag links that exist in all HTML files were gathered and saved for further analysis. Non-informative elements such as table-of-contents links and numeric anchors were removed from unique hyperlink-text pairs. In the subsequent stage, the frequency of **background** keywords in the links was analyzed to understand the most common patterns in the Edgar filings, which can be used to extract the background section of transactions. To determine the boundaries of each section, we evaluate the hyperlinks after the hyperlink related to the background section. By skipping links such as table-of-contents entries, numeric anchors, duplicates, or empty tags, we generate structured start-and-end pairs for each filing. These results are saved to enable systematic extraction of the background section of deals. Tables 3.3 and 3.4 show the frequency of such links' texts. Even though some terms, including firm names or unusual wording, made it hard to consistently find the background section, this analysis helped us come up with good regex patterns and stop-phrase lists.

3.5 Extracting Background Sections

For extracting the background section of deals, we focused on start-and-end pairs discussed in previous steps to find the relevant and candidate background sections in filings. We differentiated among href-based files (HTML files containing `<a>` tag links), no href-based files (HTML files without any tag links to focus), and raw txt files (these files are usually from the 90s and the first years of the 2000s).

3.5.1 Href-Based Extraction

Based on the detection of background section titles and section results, textual patterns were prepared to identify the background section of deals. Each HTML file was parsed, and a targeted regular-expression matcher identified anchor labels that potentially correspond to the beginning of a background segment by searching for text fragments where the keyword “background” appears near transaction-related terminology. For each identified starting anchor, a stopping anchor was extracted by locating the nearest subsequent anchor whose text or href differs from the start point. Using a custom DOM traversal, all textual content occurring between the identified start and stop anchors was extracted.

3.5.2 Non-Href Structured Extraction

For HTML files without reliance on href tags, we focused on all tags whose text length falls within a heuristically determined window (3–15 words), excluding non-informative tags. This filtered tag set is then analyzed to detect candidate section markers: a start marker must contain the term “background” along with transaction-related keywords (e.g., “merger” or “acquisition”), whereas an end marker must contain end phrases and terms such as “reasons” or “recommendation,” which indicates the end of the background section of deals. To form candidate start–end tag pairs, the DOM traversal method was used to extract the text chunks. Among all candidate texts, we select the “best” background section by applying a minimum word-count threshold and rejecting chunks that are strict subsets of others.

3.5.3 Raw Plain-Text Files Extraction

For raw texts, it is segmented into paragraphs, and only medium-length textual units (3–15 words) that plausibly correspond to headers of sections were retained. From this filtered subset, potential start markers were detected by requiring that a line contain the term “background” together with merger-related keywords (e.g., merger, acquisition, transaction). Then we search forward for the first end marker containing transition terms such as “reasons” or “recommendation,” forming candidate start-end pairs. Using these exact markers, the script performs a line-by-line extraction on the original full text to retrieve all blocks of text located between each pair. We select the longest section exceeding a minimum threshold (200 words).

3.6 Automated Extraction of Introductory Sections From Filings

During our analyses focused on extracting target and final acquirer official names, we understood that in some background sections there is no direct name provided. To resolve such issues, we focused on extracting the introductory part of these filings. In the first part of the documents, the name of the target and final acquirer can come in the letter for the stockholders to accept or reject the bid to the target company and information about the meeting of stakeholders’ time and location. Our focus for identifying the stopping phrases was the **"table of contents"** and **"notice of meeting"** phrases found at the beginning of documents. For TXT files, regular expressions were used to detect end markers and extract the relevant text. For HTML files, a recursive traversal of the DOM tree is performed to extract textual content until the stopping markers are encountered. The resulting introductory texts are saved in a structured directory hierarchy.

For each document, the extraction process depends on the file type. For TXT files, the script first searches for a Table of Contents (TOC) and extracts all text preceding it as the introductory section if the word count is between 500 and 10,000 words. If no TOC is found or the extracted section does not meet the word count requirement, it searches for a Notice of Meeting (special or annual) and extracts all text before it under the same constraints. For HTML files, the script similarly first looks for a TOC string (ignoring hyperlinks) and extracts preceding text if valid, and if that is absent or invalid, it searches for a notice of meeting string and extracts the text

before it. If a document contains neither a TOC nor a Notice, the script attempts to extract the introductory section from the beginning of the document, truncating the extracted content to approximately 8,000 tokens to ensure consistency.

3.7 Final Dataset for Analyzing the Private Phase of M&A

As shown in Figure 3.1, the majority of filings are of type S-4, followed by DEFM14A and PREM14A filings. This distribution reflects transactions' filings with detailed narratives of the merger and acquisition processes. After applying all filtering and extraction steps discussed in previous sections, the dataset comprises a total of 14,751 filings. Table 3.5 shows the token counts using the Qwen3-8B model used in the analysis.

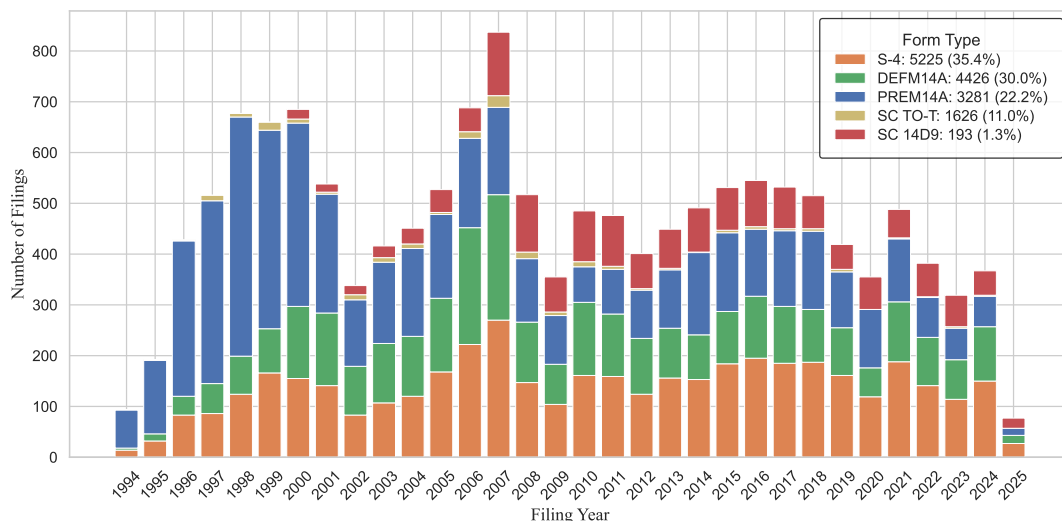


Figure 3.1 **Distribution of SEC Filings by Form Type and Year.** This figure illustrates the annual counts of filings for each form type, highlighting trends over time.

Table 3.1 EDGAR Filings: Number of Files per Format and Total Size (1994–2025)

Year	TXT	HTM	HTML	Total Files	Total Size (MB)
1994	340	0	340	680	249.39
1995	607	0	607	1214	378.44
1996	1501	0	1501	3002	1462.51
1997	2881	0	2881	5762	3245.58
1998	3759	0	3759	7518	5049.42
1999	4138	0	4177	8315	6158.16
2000	11845	83	3283	15211	3363.36
2001	89890	3401	5522	98813	14046.9
2002	66173	8775	4318	79266	12564.5
2003	146370	32511	5863	184744	32136.4
2004	94635	88592	6008	189235	38395.5
2005	71567	107796	5351	184714	39126.2
2006	37826	168356	4299	210481	40314.5
2007	8548	262592	5291	276431	48578.3
2008	2681	166335	2525	171541	38135.5
2009	2602	44824	2573	49999	19861.2
2010	6038	317531	6034	329603	63245.3
2011	7411	439065	7410	453886	110279
2012	4665	231735	4668	241068	64781.6
2013	7229	381595	7268	396092	108752
2014	5539	409167	9530	424236	89120.5
2015	5060	202852	10120	218032	57792.8
2016	3594	292197	7184	302975	60520.4
2017	3571	188568	7142	199281	46485.1
2018	2235	48405	4462	55102	17017.3
2019	1875	99942	3750	105567	44018.3
2020	1672	38920	3344	43936	17458
2021	1820	50202	3640	55662	29058.5
2022	1403	19135	2806	23344	16698.1
2023	899	18193	1798	20890	15532.6
2024	1007	19206	2014	22227	17387.2
2025	161	2013	322	2496	2166.87
Total	722,237	2,965,276	100,407	3,789,516	1,038,460 (~1 TB)

Table 3.2 Summary of Qualified Files by Year

Year	TXT	HTM	HTML	Total Files	Total Size (MB)
1994	299	0	0	299	221.92
1995	511	0	0	511	323.20
1996	1106	0	0	1106	945.07
1997	1585	0	0	1585	1556.65
1998	2060	0	0	2060	2298.88
1999	2470	0	8	2478	2681.78
2000	2863	13	18	2894	2018.38
2001	5382	250	2	5634	5029.04
2002	3456	604	0	4060	4427.46
2003	3952	1770	13	5735	9463.98
2004	4158	2779	0	6937	15990.70
2005	4377	2323	0	6700	19564.90
2006	0	2859	0	2859	4034.77
2007	0	2823	0	2823	4170.67
2008	0	3985	0	3985	6550.96
2009	0	1704	0	1704	4351.12
2010	0	4046	0	4046	8381.19
2011	0	5723	0	5723	10954.90
2012	0	3472	0	3472	7020.68
2013	0	5553	0	5553	8854.67
2014	0	3455	0	3455	5336.04
2015	0	3381	0	3381	4734.89
2016	0	3167	0	3167	5061.71
2017	0	2108	0	2108	4236.22
2018	0	1103	0	1103	2455.94
2019	0	1441	0	1441	5513.83
2020	0	1126	0	1126	3102.35
2021	0	1283	0	1283	4629.40
2022	0	1004	0	1004	3373.03
2023	0	903	0	903	3417.78
2024	0	935	0	935	3558.49
2025	0	197	0	197	678.21
Total	43676	72675	41	116392	164,645.02 (~160.7 GB)

Table 3.3 Top 30 Most Common Hyperlink-Text Pairs Containing "**Background**" Keyword in M&A Filings

No.	Link Text	Count
1	background of the merger	5633
2	background	385
3	background of the mergers	341
4	background of the transaction	261
5	background of the\n merger	226
6	background of the transactions	208
7	identity and background of filing person	194
8	item 3. identity and background of filing person.	186
9	background to the merger	165
10	item 2. identity and background of filing person.	164
11	background of the offer	128
12	company background	114
13	item 2. identity and background of filing person	97
14	item 2. identity and background of filing person	92
15	item 3. identity and background of filing person	91
16	background\n of the merger	89
17	background of the asset sale	89
18	background of the business combination	85
19	identity and background of filing person.	82
20	background and reasons for the merger	66
21	background of and reasons for the merger	63
22	industry background	57
23	background of the acquisition	56
24	item 3. identity and background of filing person	55
25	background of the offer; past contacts or negotiations	52
26	background to the arrangement	51
27	item 2. identity and background of filing person.	49
28	background of the arrangement	43
29	background to the offer	42
30	background\n of the merger	42

Table 3.4 Top 30 Most Common Ending Phrases for Background Sections in M&A Filings

No.	Link Text	Count
1	Reasons for the Merger	545
2	Recommendation of the Board	130
3	The Merger Agreement; Other Agreements	120
4	Recommendation of the Board and Reasons for the Merger	105
5	Recommendation of the Board of Directors and Reasons for the Merger	102
6	The Merger Agreement; Other Agreements.	88
7	Recommendation of the Board of Directors and Reasons for the\n Merger	85
8	Reasons for the Merger; Recommendation of the Board of Directors	77
9	Reasons for the\n Merger	61
10	Reasons for the Merger; Recommendation of Our Board of Directors	51
11	Reasons for the Merger; Recommendation of the Board	48
12	Recommendation of Our Board of Directors and Reasons for the Merger	46
13	Reasons for the Mergers	37
14	Recommendation of Our Board of Directors	37
15	Reasons for the Merger and Recommendation of Our Board of Directors	34
16	Reasons for the Merger and Recommendation of the Board of Directors	32
17	Reasons for the Asset Sale	32
18	Reasons for the Merger; Recommendation of Our\n Board of Directors	30
19	Recommendation of Our Board of Directors and Reasons for the\n Merger	26
20	Reasons for the Merger; Recommendation of the Board of\n Directors	25
21	Recommendation of the Board of Directors	24
22	THE MERGER	24
23	PDC's Reasons for the Merger	24
24	The Merger Agreement	23
25	Determination of Merger Consideration	23
26	Background of the Offer	23
27	Recommendation of the Special Committee	21
28	Recommendation of our Board of Directors	21
29	Reasons for the Merger and Recommendation of our Board of Directors	20
30	Reasons for the Merger; Recommendation of the\n Board of Directors	20

Table 3.5 Token Count Distribution in Introductory and Background Sections

Statistic	Background	Intro
Min	265	794
P01	528	1,182
P05	1,070	1,650
P25 (Q1)	2,563	2,744
Mean	7,933	4,376
P75 (Q3)	9,052	5,367
P90	14,406	7,910
P99	56,433	11,839
Max	515,293	82,956
Std. Dev.	16,771	2,747

4. CONSTRUCTION OF THE PRIVATE-PHASE M&A DATASET

This chapter describes the construction of a novel dataset based on SEC filings, especially focused on the private phase of mergers and acquisitions (*Background Section*). We extract deal-relevant information, including transaction counterparties, private-phase events, and bid terms. The extraction framework combines large language model (LLM) prompting with language processing methods and manual annotation to transform unstructured textual disclosures into structured data.

Specifically, we use an LLM-based multistage framework to identify transaction parties and private-phase deal events from the background introductory sections of filings. To extract numeric bid prices and transaction terms, we complement LLM extraction with rule-based natural language processing using spaCy (`en_core_web_trf` transformer model¹), which allows for precise identification and normalization of monetary expressions. Finally, we construct a ground-truth benchmark through manual annotation and implement entity- and event-level validation procedures to assess extraction accuracy. This hybrid approach enables the systematic transformation of unstructured narrative disclosures into a structured dataset suitable for empirical analysis.

4.1 Large Language-Based Inference Pipeline

This section discusses the LLM-based methodology used to analyze the EDGAR SEC filings and extract valuable information. Figure 4.1 shows clearly the framework for extracting information from extracted sections using an LLM. This multi-stage framework was necessary, as each step extracted information that was essential for

1. https://huggingface.co/spacy/en_core_web_trf

future steps. The black arrows show sections from which JSON schema information was utilized in prompts. The blue arrows correspond to resulting JSON schemas as LLM output.

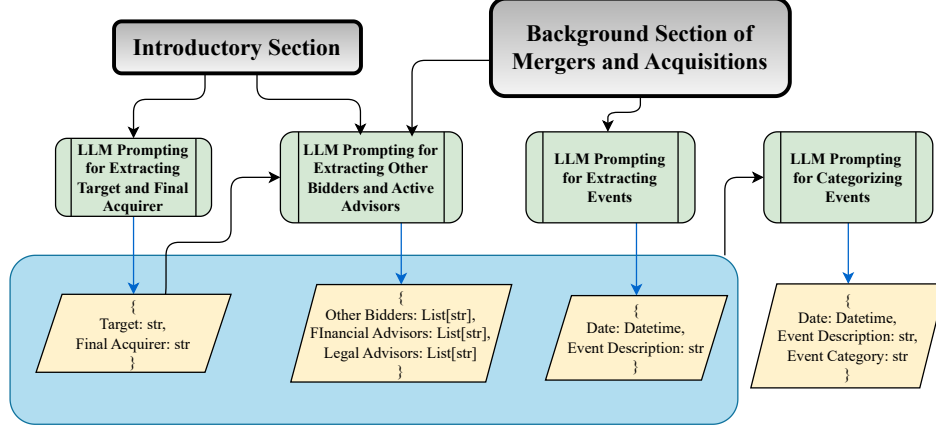


Figure 4.1 Workflow of the large language model-based inference pipeline, illustrating the chained extraction of transaction parties, external advisors, and private-phase negotiation events from SEC EDGAR filings.

4.1.1 Model Selection and Inference Configuration

This study employs the Qwen3-8B² causal large language model, a decoder-only transformer architecture released by the Qwen team and distributed via Hugging Face. The model contains approximately 8.2 billion parameters, with a context length of 32,768 and 131,072 tokens using YaRN.

The selection of Qwen3-8B is motivated by three considerations. First, the model exhibits strong instruction-following behavior and reliable performance in structured information extraction tasks. This ability enabled us to extract JSON schemas in each prompting stage. Second, the model had a high accuracy report for information extraction from SEC filings, both for parties involved and later on for event-related categorization. Third, the model supports an explicit reasoning-enabled inference configuration, which is well suited to the long narrative structure of SEC filings and the understanding of the relations of parties and events.

2. <https://huggingface.co/Qwen/Qwen3-8B>

All inferences were conducted with the model’s *thinking* (reasoning) mode enabled. This configuration allows the model to internally perform multi-step logical reasoning before generating its final output. Inference was performed using fixed prompt templates and constrained output schemas, implemented via the `pydantic` module, to enhance reproducibility and limit variation across runs.

The system prompt *“You are a helpful AI assistant for Mergers and Acquisitions.”* was used consistently across all prompting stages. The same model configuration was applied uniformly to all extraction tasks, with the only variation being the `max_new_tokens` parameter. For each inference step, both `thinking_content` and `content` outputs were generated; however, only the main `content` output was retained for downstream analyses. Additional details regarding prompt structures are provided in Appendix A.

4.1.2 Identification of Transaction Parties

This section describes the extraction of transaction-related entities from SEC filings, including the target firm, the final acquirer, other potential bidders who were unsuccessful in the end, and external financial and legal advisors.

4.1.2.1 Identification of the Target and Final Acquirer

The first attempt to extract transaction parties focused on identifying the target and the final acquirer firms from the **introductory section** of SEC filings. The introductory section was selected rather than the background section because, in many filings, official company names are not consistently stated in the background narrative. Instead, the background section sometimes refers to shell entities or special purpose vehicles (SPVs) created solely to execute the transaction, which act as intermediaries on behalf of parent companies. In contrast, the introductory portion of the filing typically provides the full legal names of the transaction parties and defines standardized references (e.g., “the Company,” “the Offeror,” “the Purchaser”), which are used throughout subsequent sections of the document.

To ensure structured and consistent extraction, the inference prompt explicitly re-

stricted the model to the introductory section and required the output to follow a predefined JSON schema specifying the target and final acquirer. The prompt included formal definitions of each role and explicit rules instructing the model to ignore transaction vehicles, subsidiaries, committees, and advisors and instead identify the ultimate parent company controlling the acquisition. This design was necessary to align the extracted firm names with the SDC dataset, which records official legal entity names rather than subsidiary or SPV identifiers.

The generation length was constrained by setting the `max_new_tokens` parameter to 512, which was sufficient for producing the structured JSON output while preventing unnecessary narrative text. The strict output format requirement (JSON-only, no explanations) was imposed to facilitate downstream parsing and automated matching with structured deal data.

4.1.2.2 Identification of Other Bidders and External Advisors

In the second step of entity extraction, the **background section**, the **introductory section**, and the previously extracted **target** and **final acquirer** names were incorporated into the inference prompt. The primary motivation for jointly extracting other bidders and external financial and legal advisors was to allow the model to explicitly differentiate among entity roles and reduce misclassification.

The prompt explicitly instructed the model to use the introductory section only to identify the target and final acquirer, while restricting the extraction of other bidders and external advisors to the background section. Formal role definitions and exclusion rules were provided to prevent confusion between bidders, advisors, subsidiaries, and transaction vehicles. In particular, shell companies and special purpose vehicles (SPVs) were explicitly defined and excluded from all categories unless they represented the true parent entity, ensuring consistency with structured deal databases such as SDC, which record ultimate legal entities rather than transaction vehicles.

A structured JSON schema was imposed as the output format, requiring separate lists for other bidders, financial advisors, and legal advisors. The prompt further enforced strict rules regarding de-duplication, preservation of order of appearance, and omission of ambiguous entities, favoring precision over recall when role assignment was uncertain.

The `max_new_tokens` parameter was set to 10,000 for this extraction stage. The larger token budget was required due to the greater linguistic complexity of the background section and the need for the model to reason over multiple entities, role definitions, and exclusion constraints while producing structured outputs. The introductory section was included despite the availability of target and final acquirer names from the first stage, as it provided essential contextual information for identifying and excluding shells and SPVs that appeared in the background narrative but did not represent the ultimate transaction parties.

4.1.3 Private-Phase Event Identification and Analysis

Beyond role extraction in previous sections, one of the significant characteristics of the background section of the filings is its narrative style with varied formats of dates. This section focuses on extracting events and categorizing them in the M&A context.

4.1.3.1 Event Extraction from Background Section

To extract structured transactional events from the *Background of the Transaction* sections, each background document was segmented into semantically coherent text chunks using sentence-level tokenization. This chunking strategy ensured that temporally related interactions were processed together and that long disclosures could be analyzed without loss of relevant information. For each chunk, the large language model (LLM) extracted chronological transactional events and their associated temporal information, producing a JSON array of event objects for each transaction.

Each background document was first split into sentence-based chunks using the NLTK package³, with each chunk constrained to a maximum of 1,000 model tokens. To avoid generating context-poor final segments, any trailing chunk containing fewer than 60% of the token limit was merged with the preceding chunk. This merging rule ensured sufficient contextual continuity for accurate identification of temporally linked events. During inference, the `max_new_tokens` parameter was set to 7,000 per chunk to accommodate the more extensive reasoning required for identifying

3. <https://github.com/nltk/nltk>

detailed chronological events and generating structured outputs.

The inference prompt instructed the model to extract every interaction or event in the order of appearance and to return one JSON object per event following a predefined **ChronoEntry** schema. The schema required a standardized temporal field and a concise event description, with explicit precedence rules for temporal expressions ranging from exact dates to approximate or unspecified time references. The prompt further enforced strict output constraints, including JSON-only responses, chronological ordering, and mandatory inclusion of events meeting predefined criteria such as engagements, agreements, expressions of intent, and financial terms.

To improve temporal coherence and reduce fragmentation, the prompt introduced explicit rules for grouping micro-events into coherent episodes when they occurred in close textual proximity, involved the same parties, and represented sequential steps of a single interaction. At the same time, the model was instructed to avoid reordering events, omit purely narrative sentences, and explicitly mark missing information using bracketed placeholders. These constraints were designed to facilitate downstream parsing, ensure temporal consistency, and enable systematic alignment with structured M&A process data.

4.1.3.2 Event Categorization

In the final stage of the prompting pipeline, we perform event categorization. The inputs to the model consist of the structured chronological events extracted from the background section, along with the identified **target**, **final acquirer**, **other bidders**, and **financial advisors**. Each event is assigned to exactly one of five mutually exclusive categories: *Target Financial Advisor Hiring*, *Confidentiality Agreement*, *Price-Specific Offers by Bidders*, *Deal Announcement*, and *Other*.

To control generation length while respecting the model’s maximum context window, the `max_new_tokens` parameter was determined dynamically for each prompt based on the input token length. When the prompt contained 10,000 tokens or fewer, the generation budget was set to three times the input length, ensuring sufficient capacity for internal reasoning and structured classification while avoiding premature truncation. For prompts exceeding this threshold, `max_new_tokens` was computed as the remaining available context after subtracting the input length and a fixed 500-token safety buffer from the model’s maximum context size (40,960

tokens). This adaptive strategy prevented context overflow while still allowing the model to complete classification and output generation.

The inference prompt required the model to return a JSON array of categorized event objects following a predefined schema, with each object containing the original date, the original event description, and a single category label. The prompt enforced that the date and description fields be returned verbatim, preventing the model from paraphrasing or altering extracted events. Category definitions were explicitly provided, along with process guidelines instructing the model to match each event to the most appropriate category based on the roles of the target, final acquirer, bidders, and advisors. Events that did not meet any predefined criteria were assigned to the residual *Other* category.

Illustrative examples from unrelated transactions were included in the prompt to demonstrate correct categorization patterns and reduce ambiguity in role assignment. These examples covered advisor hiring events, confidentiality agreements, price-specific bid proposals, and public deal announcements. The model was instructed to process events in chronological order and assign exactly one category per event, thereby producing a structured and mutually exclusive event taxonomy suitable for downstream empirical analysis.

4.2 Annotation Procedure and Ground-Truth Construction

To evaluate the accuracy of the large language model-based extraction pipeline, a manually curated ground-truth dataset was constructed through a structured annotation procedure. A random sample of transaction filings was selected and independently reviewed to identify key deal entities and private-phase events disclosed in the *Background of the Transaction* sections.

The annotation process targets both entity-level and event-level information. First part of the annotation focused on entity-level information, such as the target firm, final acquirer, other bidders, and financial and legal advisors. For a subset of 50 transactions, all relevant deal roles are fully annotated.

In addition, for a subset of 20 transactions drawn from this sample, comprising a total of 1,033 extracted chronological events, each event is manually classified into

the standardized event categories employed in the empirical analysis.

4.2.1 Entity-Level Exact and Fuzzy Matching Accuracy

For entity matching, ground-truth labels were created by recording the names of the target firm and final acquirer, as well as all explicitly mentioned other bidders and advisors. Exact string matching was used to evaluate target and final acquirer identification.

Table 4.1 reports exact match ratios for the two primary deal entities across 50 annotated transactions. The results indicate near-perfect identification accuracy, with a match ratio of 98.08% for target firms and 100% for final acquirers.

Table 4.1 Exact Match Accuracy for Target and Final Acquirer, based on 50 deals.

Entity Type	Match Ratio (%)
Target Firm	98.08
Final Acquirer	100.00

For other bidders, the annotation focused exclusively on explicitly named entities or clearly defined placeholders (e.g., “Company A,” “Company B,” “the Company”). Observations in which the *Background of the Transaction* section disclosed only an aggregate number of bidders, without identifying individual parties, were excluded from annotation.

For financial and legal advisors, the annotation was restricted to the official names of external advisory firms. Generic references or placeholders were not permitted for external advisors’ annotations, and only explicitly identified third-party advisors were recorded.

Two entity strings are considered a match if, after normalization, they are identical, one string is a substring of the other, or their fuzzy string similarity score (based on the Levenshtein ratio) exceeds a predefined threshold of 85.

Table 4.2 presents summary statistics for these metrics. Across all fields, median scores equal one, indicating that in at least half of the annotated deals, model outputs exactly match or fully cover the human-labeled entity sets.

Annotated Coverage. Annotated coverage captures the extent to which manually annotated entities are recovered by the LLM extraction pipeline. For a given

transaction document, let A denote the set of annotated entities (e.g., other bidders or external advisors) and B the set of entities extracted by the LLM. An annotated entity $a \in A$ is considered covered if it can be matched to at least one entity $b \in B$ according to the string-matching logic defined above. Annotated coverage is defined as

$$\text{Annotated Coverage} = \frac{|A \cap B|}{|A|},$$

where $|A \cap B|$ denotes the number of annotated entities that are successfully matched under one-to-one matching.

LLM Coverage. LLM coverage measures the proportion of entities extracted by the language model that correspond to manually annotated ground-truth entities. Using the same notation, an extracted entity $b \in B$ is considered covered if it matches at least one annotated entity $a \in A$ according to the same matching logic. LLM coverage is defined as

$$\text{LLM Coverage} = \frac{|B \cap A|}{|B|},$$

where each extracted entity can be matched to at most one annotated entity.

Fuzzy Jaccard Similarity. Fuzzy Jaccard similarity generalizes the classical Jaccard index to measure agreement between annotated and LLM-extracted entity lists under approximate string matching. Let A and B denote the sets of annotated and extracted entities, respectively. The intersection $|A \cap B|$ is defined as the number of one-to-one matched entity pairs identified according to the matching logic described above. The fuzzy Jaccard similarity is then given by

$$J_{\text{fuzzy}}(A, B) = \frac{|A \cap B|}{|A| + |B| - |A \cap B|}.$$

4.2.2 Event-Level Categorization Accuracy

Ground-truth event labels were constructed by manually assigning each extracted chronological event to one of five mutually exclusive private-phase event categories: confidentiality agreements, target financial advisor hiring, price-specific offers by bidders, deal announcement, and a residual *Other* category. Annotation was performed at the event level, preserving the original chronological ordering of the background section.

Table 4.2 Summary Statistics for LLM-Based Entity Extraction Performance (50 Deals)

Metric	Field	Mean	Min	P25	Median	P75	Max
3*Fuzzy Jaccard	Other Bidders	0.87	0.00	1.00	1.00	1.00	1.00
	Financial Advisors	0.90	0.00	1.00	1.00	1.00	1.00
	Legal Advisors	0.83	0.00	0.79	1.00	1.00	1.00
3*Annotated Coverage	Other Bidders	0.95	0.00	1.00	1.00	1.00	1.00
	Financial Advisors	0.95	0.33	1.00	1.00	1.00	1.00
	Legal Advisors	0.85	0.00	0.96	1.00	1.00	1.00
3*LLM Coverage	Other Bidders	0.92	0.00	1.00	1.00	1.00	1.00
	Financial Advisors	0.96	0.00	1.00	1.00	1.00	1.00
	Legal Advisors	0.99	0.67	1.00	1.00	1.00	1.00

Notes: This table reports summary statistics for entity extraction performance across annotated deal participants. Metrics include fuzzy Jaccard similarity between LLM and human annotations, annotated coverage (fraction of annotated entities recovered by the LLM), and LLM coverage (fraction of LLM-extracted entities appearing in annotations). The sample consists of 50 randomly selected deals.

Table 4.3 reports category-level precision, recall, and F1 scores based on 20 manually annotated transactions comprising 1,033 events. The results demonstrate consistently high recall across all event categories, indicating that the pipeline captures all relevant categories without missing any. The relatively lower precision observed for the *Target Financial Advisor Hiring* category primarily reflects the inclusion of surrounding explanatory events that provide contextual information but fall outside the strict category definition present in the prompt. In general, the model tends to classify them as relevant to the category rather than assigning them to the residual *Other* category.

Table 4.3 Category-Level Performance for Event Classification

Event Category	Precision	Recall	F1	Support
Target Financial Advisor Hiring	0.49	1.00	0.65	16
Confidentiality Agreement	0.73	1.00	0.85	22
Price-Specific Offers by Bidders	0.86	0.95	0.90	78
Deal Announcement	0.68	1.00	0.81	17
Other	1.00	0.95	0.97	896
Macro Avg.	0.75	0.98	0.84	–
Weighted Avg.	0.95	0.95	0.95	1,033

Notes: This table reports precision, recall, and F1 scores for event classification using 1,033 annotated events from 20 randomly selected deals. Macro averages weight each category equally, whereas weighted averages are weighted by category support.

4.3 Matching With the SDC Platinum Dataset

To link the EDGAR filings with transactions recorded in the SDC Platinum database, a structured matching procedure that combines fuzzy name matching with deal-date validation is used. The objective of this matching is to identify the corresponding SDC transaction for each EDGAR filing.

First, the SDC dataset was filtered according to the following criteria. The sample was restricted to transactions in which either the target or the acquirer was located in the United States. Only deals classified as completed, pending, or withdrawn were retained. Transactions categorized as buyback, acquisition of certain assets, acquisition, exchange offer, or recapitalization were excluded. Observations in which the acquirer and target were the same entity were removed. Finally, multiple records corresponding to different stages of the same underlying transaction, as identified by identical target and acquirer names, were grouped into a single observation. Table 4.4 shows the distribution of the filtered SDC dataset based on form of deal and deal status.

Table 4.4 Distribution of Deal Characteristics in the Filtered SDC Dataset

Panel A: Form of the Deal	
Acquisition of Assets	34,621
Merger	21,697
Acquisition of Partial Interest	13,811
Acquisition of Majority Interest	2,564
Acquisition of Remaining Interest	1,008
Panel B: Deal Status	
Completed	66,778
Withdrawn	3,486
Pending	3,437

All relevant target- and acquirer-related names are standardized prior to matching. For EDGAR, target and acquirer names extracted in the first step from the introductory part of filings are converted to lowercase, non-alphanumeric characters are removed, and excess whitespace is collapsed. In the SDC data, normalization is applied not only to the primary target and acquirer names but also to their immediate and ultimate parent names. This allows the procedure to account for transactions conducted through subsidiaries or holding companies.

Candidate SDC matches for a given EDGAR record are identified using fuzzy string similarity. For each EDGAR file, token-set similarity scores are computed between

the **normalized EDGAR target name** and each of the normalized **SDC target-related fields**, and separately between the **normalized EDGAR acquirer name** and the corresponding **SDC acquirer-related fields**. For each SDC deal, the maximum similarity score across the target-related fields and the maximum score across the acquirer-related fields are retained. These two scores are then averaged to form a combined similarity measure. Only SDC deals with a combined similarity score above a predefined threshold (**80 out of 100**) are retained as potential candidates.

The final match selection proceeds in two sequential phases. In the first phase, candidate SDC deals are evaluated within a deal-specific time window. For each candidate, the window is defined using the SDC **announcement date** as the start date and either the **effective date (for completed deals)**, the **withdrawal date (for withdrawn deals)**, or the announcement date itself (for pending deals) as the end date. An EDGAR observation is considered a valid match if its event date falls within this window. When multiple candidates satisfy this condition, the deal with the highest combined similarity score is selected. Once a match is established, the corresponding SDC observation is removed from the candidate pool to enforce a one-to-one matching structure.

In the second phase, remaining unmatched EDGAR observations are compared against the remaining SDC deals using a closest-date criterion. This phase applies stricter requirements, including a higher similarity threshold (**90 out of 100**) and a maximum allowable date difference (**365 days**) between the EDGAR date filed and the SDC announcement date. Among the candidates that satisfy these constraints, the SDC deal with the smallest absolute date difference is selected. As in the first phase, matched SDC records are immediately removed to prevent multiple assignments.

Following the matching procedure between the EDGAR-based sample and the filtered SDC Platinum dataset, a total of **6,207** EDGAR observations were successfully matched with SDC transactions. Table 4.5 shows the matching percentage for EDGAR and SDC datasets.

Table 4.5 Matching Outcomes Between EDGAR and SDC

Dataset	Total Deals	Matched (%)	Unmatched (%)
EDGAR	14,751	42.1	57.9
SDC Platinum	73,701	8.4	91.6

Notes: This table reports the fraction of deals successfully matched across EDGAR filings and the SDC Platinum database. Percentages are computed relative to the total number of deals in each dataset.

4.4 Extraction of Price-Specific Bidder Offer Prices Using SpaCy

To extract price information from the private-phase M&A narratives, we implement a hybrid natural language processing pipeline that combines transformer-based named entity recognition (NER) with rule-based numerical parsing. The focus is on events categorized as *Price Specific Offers by Bidders*, which correspond to explicit bidder proposals containing monetary bid values.

We use the spaCy `en_core_web_trf` transformer model for monetary entity recognition, identifying text spans labeled as `MONEY`. Because transformer NER models do not return normalized numeric values, we implement a custom rule-based parser using regular expressions to extract and standardize numerical price mentions. The parser identifies numeric expressions with optional scale modifiers (e.g., “million”, “billion”) and converts them into floating-point dollar values. Duplicate values within each event are removed to avoid overcounting repeated mentions, but not across deal events. Finally, event-level price lists are aggregated to the deal level. For each deal, we construct a consolidated list of all bid prices observed during the private negotiation phase.

5. EMPIRICAL RESULTS

This chapter presents the empirical findings from the analysis of merger and acquisition transactions using both EDGAR textual filings and structured data from the SDC database. The results are organized around several key dimensions: the distribution and temporal patterns of deal parties, the dynamics and transitions across events and their corresponding predefined categories, and a detailed examination of price-specific offer events and price revisions derived from those events. We also provide a comparison of the matched SDC-EDGAR deals and their associated financial and legal advisors.

5.1 Other Bidders and External Advisors: Distribution and Trends

This section analyzes the distribution of *competing bidders* and *external financial and legal advisors* across transactions, as well as temporal changes in their frequency over time. Figure 5.1 shows that many transactions have no other bidders in the deal process, with a highly right-skewed distribution that declines sharply as the number of competing bidders increases. The figure also indicates that most deals involve a single financial advisor and two legal advisors. For many deals, a zero number of legal and financial advisors indicates that there was no explicit official name for external advisors in the background section of SEC filings.

Table 5.1 reports the twenty most frequently identified participants in M&A transactions. Panel A presents the most frequently identified *other bidders*. The distribution is dominated by generic placeholder names such as *Company A*, *Party A*, *Party B*, and *Company B*, which appear more often due to confidentiality reasons. Panel B reports the twenty most frequently reported *financial advisors*. The results

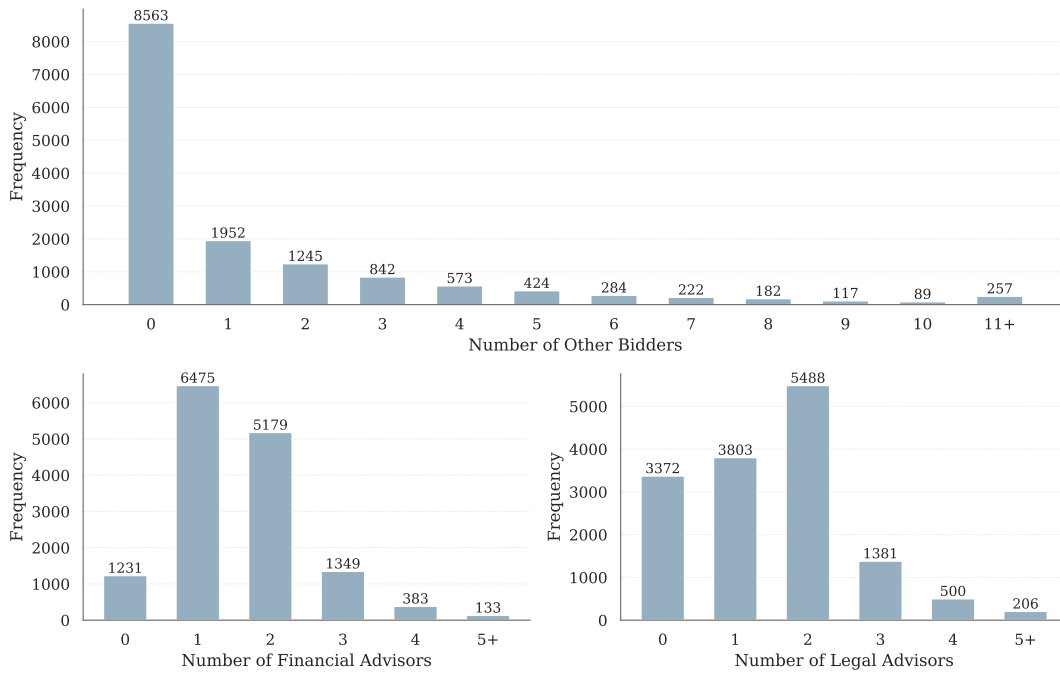


Figure 5.1 Distribution of the number of other bidders (top panel) and the number of financial and legal advisors (bottom panels) involved in M&A transactions.

reveal a high degree of concentration among a small set of major investment banks. Multiple entries corresponding to the same institution under different legal entity names appear among the most frequent advisors, reflecting variation in disclosure styles among deals. Panel C summarizes the twenty most frequently reported *legal advisors*. The distribution is similarly concentrated, with a limited group of large U.S. law firms appearing in several hundred transactions.

Figure 5.2 shows a clear upward trend in the mean and median number of other bidders, legal advisors, and financial advisors over time. For financial advisors, the interquartile range is largely concentrated between one and two advisors, while for legal advisors it spans from zero to approximately 2.5 advisors. The mean number of other bidders exceeds one beginning in 2006, while the median surpasses one for the first time in 2009.

Figure 5.3 exhibits the yearly evolution of Pearson and Spearman correlation coefficients in M&A transactions across the three panels. Panel A demonstrates that the correlation between the presence of other bidders and the involvement of financial advisors is relatively weak, generally fluctuating between 0.00 and 0.25 throughout the observed period from 1994 to 2025. Similarly, Panel B shows that the relationship between other bidders and legal advisors is nearly nonexistent, with correlation coefficients near to the zero line. In contrast, Panel C reveals a much stronger, con-

Table 5.1 Key Participants in M&A Transactions

Panel A: Top 20 Most Frequently Identified Other Bidders

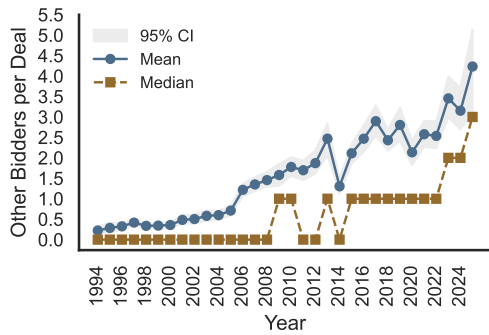
Rank 1–7		Rank 8–14		Rank 15–20	
Bidder	Freq.	Bidder	Freq.	Bidder	Freq.
Company A	1,525	Party E	572	Bidder A	201
Party A	1,442	Company D	559	Party I	186
Party B	1,295	Party F	440	Bidder B	181
Company B	1,221	Company E	376	Third Party	177
Party C	988	Party G	339	Company G	176
Company C	821	Company F	258	Party J	137
Party D	785	Party H	243		

Panel B: Top 20 Most Frequently Reported Financial Advisors

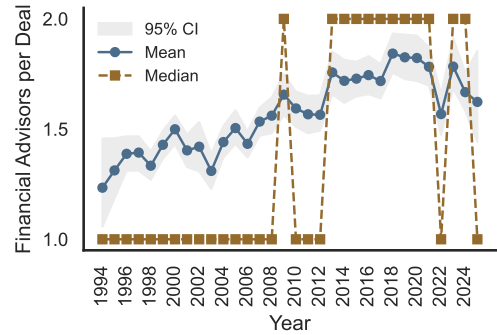
Rank 1–10		Rank 11–20	
Financial Advisor	Freq.	Financial Advisor	Freq.
Goldman Sachs	914	Evercore	234
Morgan Stanley	702	Lehman Brothers	234
Goldman Sachs & Co.	536	Morgan Stanley & Co. LLC	226
J.P. Morgan	483	Lazard	223
BofA Merrill Lynch	323	Merrill Lynch	218
Morgan Stanley & Co. Incorporated	285	Citigroup Global Markets Inc.	213
J.P. Morgan Securities LLC	269	Keefe, Bruyette & Woods, Inc.	211
KBW	251	Sandler O'Neill	204
Credit Suisse	249	Citi	186
Houlihan Lokey	244	Centerview Partners LLC	181

Panel C: Top 20 Most Frequently Reported Legal Advisors

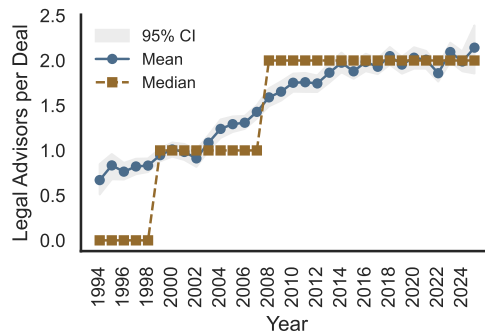
Legal Advisor	Freq.
Skadden, Arps, Slate, Meagher & Flom LLP	838
Latham & Watkins LLP	694
Wachtell, Lipton, Rosen & Katz	584
Kirkland & Ellis LLP	449
Sullivan & Cromwell LLP	420
Simpson Thacher & Bartlett LLP	323
Goodwin Procter LLP	275
Cravath, Swaine & Moore LLP	271
Weil, Gotshal & Manges LLP	270
Jones Day	266
Davis Polk & Wardwell LLP	228
Sidley Austin LLP	220
Wilson Sonsini Goodrich & Rosati, P.C.	199
Cooley LLP	193
Paul, Weiss, Rifkind, Wharton & Garrison LLP	186
Gibson, Dunn & Crutcher LLP	175
Baker Botts L.L.P.	168
Hogan Lovells US LLP	166
Ropes & Gray LLP	164
Morrison & Foerster LLP	160



Panel A



Panel B



Panel C

Figure 5.2 Yearly trends in the number of other bidders and advisor involvement in M&A transactions. Panels A–C report the yearly mean and median number of other bidders, financial advisors, and legal advisors per transaction, respectively. Shaded areas represent the interquartile range (5th–95th percentiles) obtained from bootstrap resampling of deals within each year.

sistently positive relationship between the presence of financial advisors and legal advisors, where correlation values rise from initial lows to largely stabilize between 0.40 and 0.60 for the majority of the timeline.

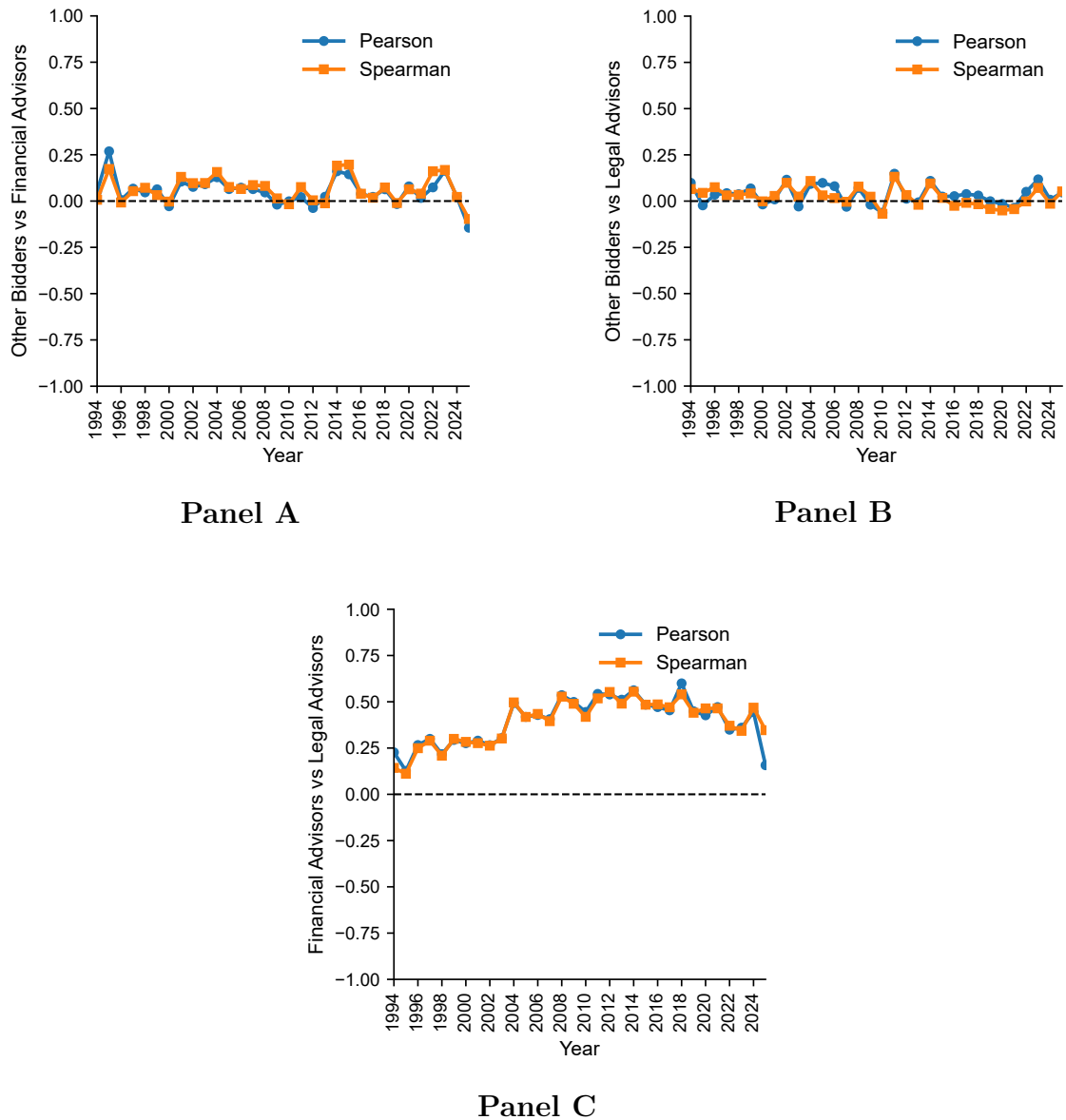


Figure 5.3 Yearly evolution of correlations between bidder competition and advisor involvement in M&A transactions. Panels A-C report Pearson and Spearman correlation coefficients between (A) other bidders and financial advisors, (B) other bidders and legal advisors, and (C) financial advisors and legal advisors.

5.2 Event Timing, Categorization, and Frequency in M&A Transactions

This section analyzes the temporal sequencing of transaction events, focusing on the ordering and duration between them in the deal process. It also presents a categorical analysis of events, their frequency, and their co-occurrence across deals.

Figure 5.4 demonstrates a shift toward more frequent transaction events over time. The left panel shows a progressive leftward shift in the distribution of log inter-event gaps, indicating that recorded events occurred closer together in recent years. The right-hand panels document this trend across three metrics: while the mean and median number of events per deal and total deal span days increased, the inter-event gap duration declined over the period.

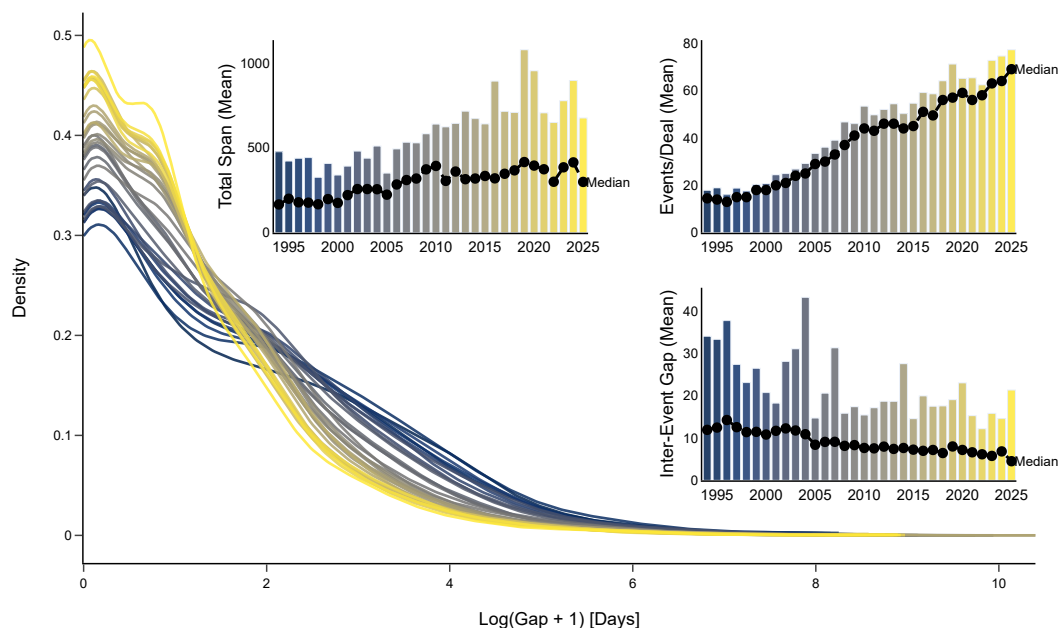


Figure 5.4 Kernel density estimates of log-transformed inter-event gaps by year, with insets reporting deal-level yearly means and medians for event counts and inter-event lags and total span days.

Figure 5.5 displays the positional distribution of key event categories along the normalized deal sequence, scaled from 0 to 100 percent within each transaction. *Target Financial Advisor Hiring* events are concentrated early in the sequence, preceding most other event categories. *Confidentiality Agreement* events follow, with an additional concentration later in the sequence, indicating multiple agreements executed at different stages. *Price-Specific Offer* events are primarily observed in the latter half of the event sequence, while *Deal Announcement* events are concentrated at the

final positions.

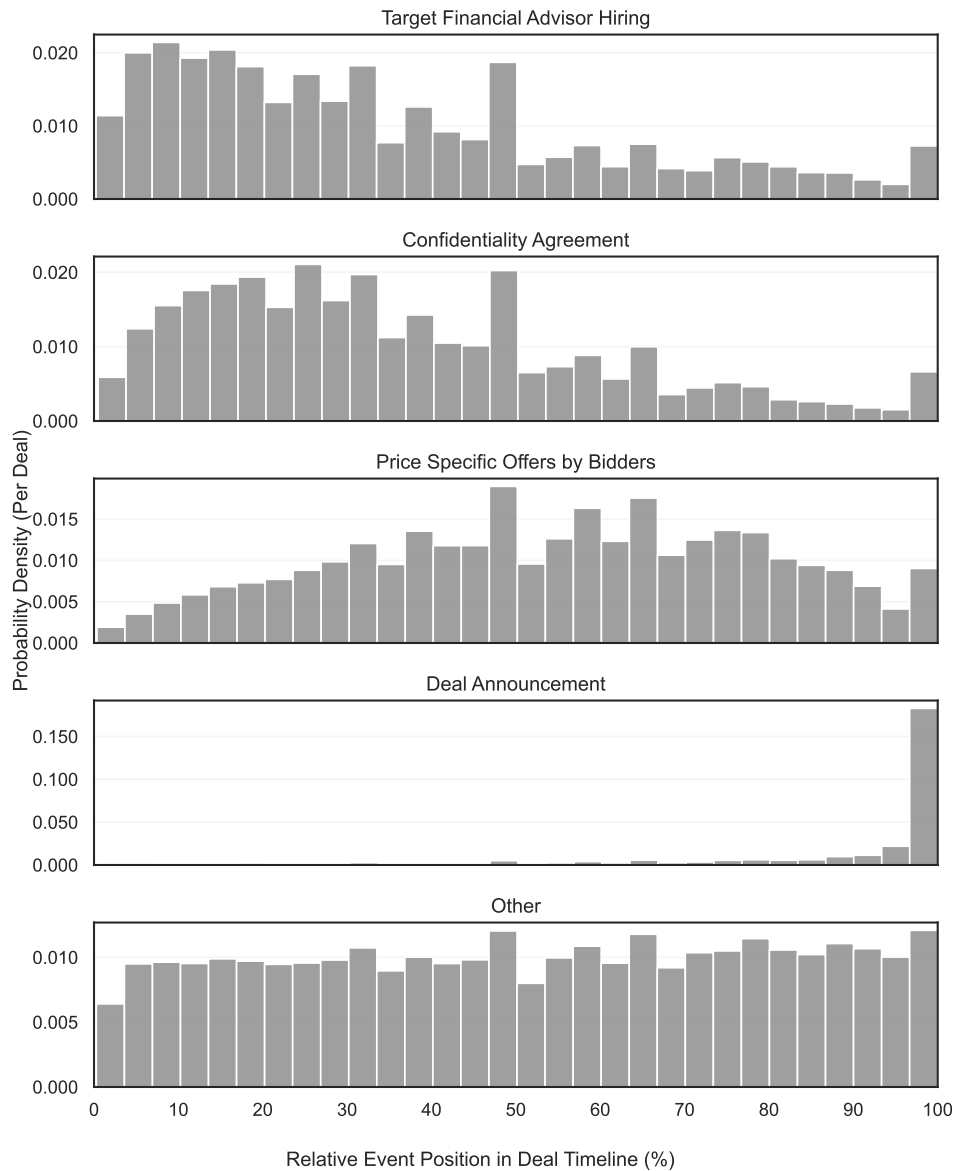
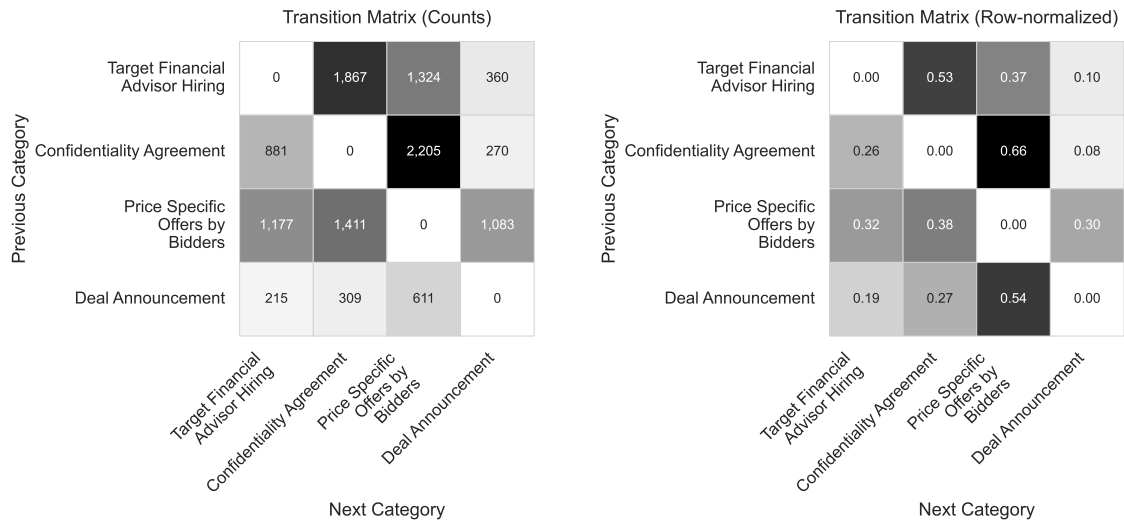


Figure 5.5 Distribution of event categories along the normalized deal event sequence. Event positions are scaled to 0–100 percent within each transaction, and panels report kernel density estimates of positional frequencies.

Figure 5.6 illustrates the sequential structure of event categories using first-order transition matrices, analogous to the positional patterns shown in Figure 5.5. Panel A reports raw transition counts between consecutive event categories, whereas Panel B presents the corresponding row-normalized transition probabilities. Target financial advisor hiring events are most frequently followed by confidentiality agreement events, with a transition probability of 0.53. Confidentiality agreement events are, in turn, most commonly followed by price-specific offer events, with a transition probability of 0.66.



Panel A: Transition counts

Panel B: Transition probabilities

Figure 5.6 First-order transition matrices for event category sequences. Panel A reports raw transition counts between consecutive event categories, while Panel B reports row-normalized transition probabilities. Consecutive repetitions of the same event category are collapsed into a single event to avoid inflating self-transitions and to focus on meaningful category changes.

Figure 5.7 reports the relative timing of price-specific offers by bidders (OFFER), confidentiality agreements (CA), and target financial advisor hiring (FA_HIRE) relative to the deal announcement date. The figure shows that target financial advisor hiring and confidentiality agreements are distributed over a long pre-announcement window, while bidder price-specific offers are concentrated close to the announcement date.

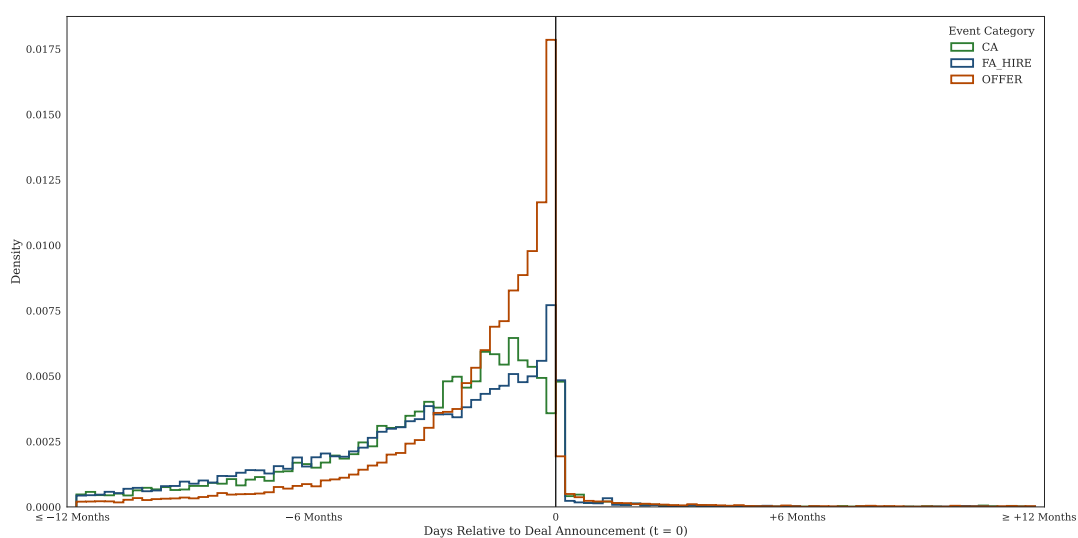


Figure 5.7 Timing of Private-Phase M&A Events Relative to Deal Announcement. Hollow histograms show the distribution of private-phase event dates in days relative to the announcement date ($t = 0$), using seven-day bins and aggregated tails beyond ± 12 months. The deal announcement date is determined as the latest event date labeled “Deal Announcement” for each transaction; if no such label exists, the last available valid event date in the original chronological order is used.

5.3 SDC and Matched EDGAR-SDC Dataset Analysis

In this section, we analyze the deal-level filtered SDC dataset for mergers and acquisitions and examine the reporting of legal and financial advisors in matched deals between the EDGAR and SDC databases. Figure 5.8 shows that for major deals like mergers and acquisitions of majority interest, there is an upward trend, whereas for other deal forms, the number of legal and financial advisors is low and the variation over time is not equivalent to major deals. Figure 5.9 demonstrates that in complicated withdrawn deals, the number of legal and financial advisors increased rapidly over the years.

As shown in Figure 5.10, advisor information is systematically underreported in EDGAR filings relative to the SDC database for both legal and financial advisors. The discrepancy is more pronounced for legal advisors than for financial advisors. For example, 19.3 percent of deals are reported in SDC as involving three financial advisors, whereas EDGAR filings report only two financial advisors for these same transactions. In contrast, the variation in reported counts is larger for legal advisors.

Figure 5.11 reports the evolution of the average number of external financial and legal advisors per deal in the matched SDC-EDGAR deals. Both datasets exhibit a clear upward trend in advisor involvement over time. For financial advisors, the SDC series shows a steeper upward slope than EDGAR, indicating that SDC systematically records a larger growth in advisor participation. A similar pattern emerges for legal advisors, where both datasets display strong positive time trends.

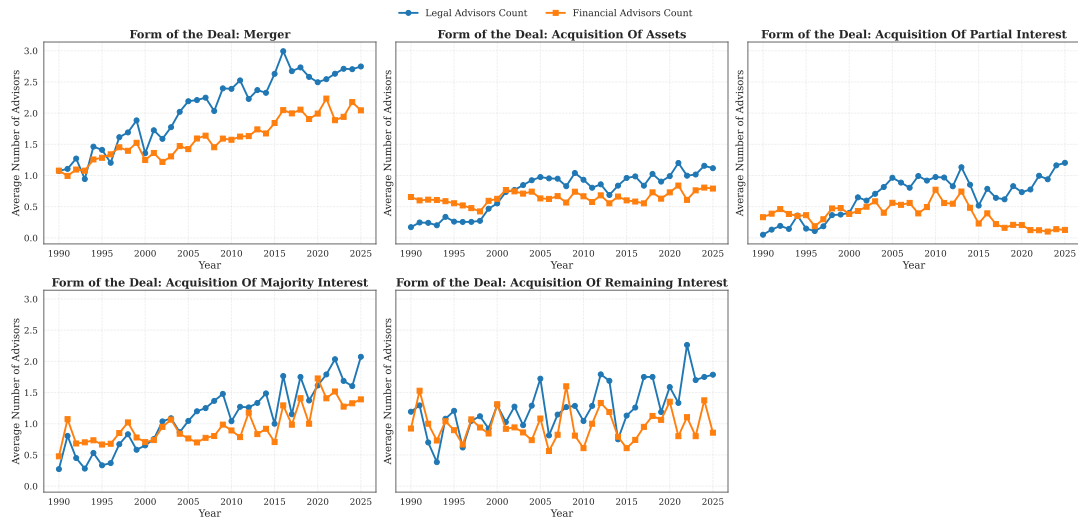


Figure 5.8 Average Number of Financial and Legal Advisors by Deal Form

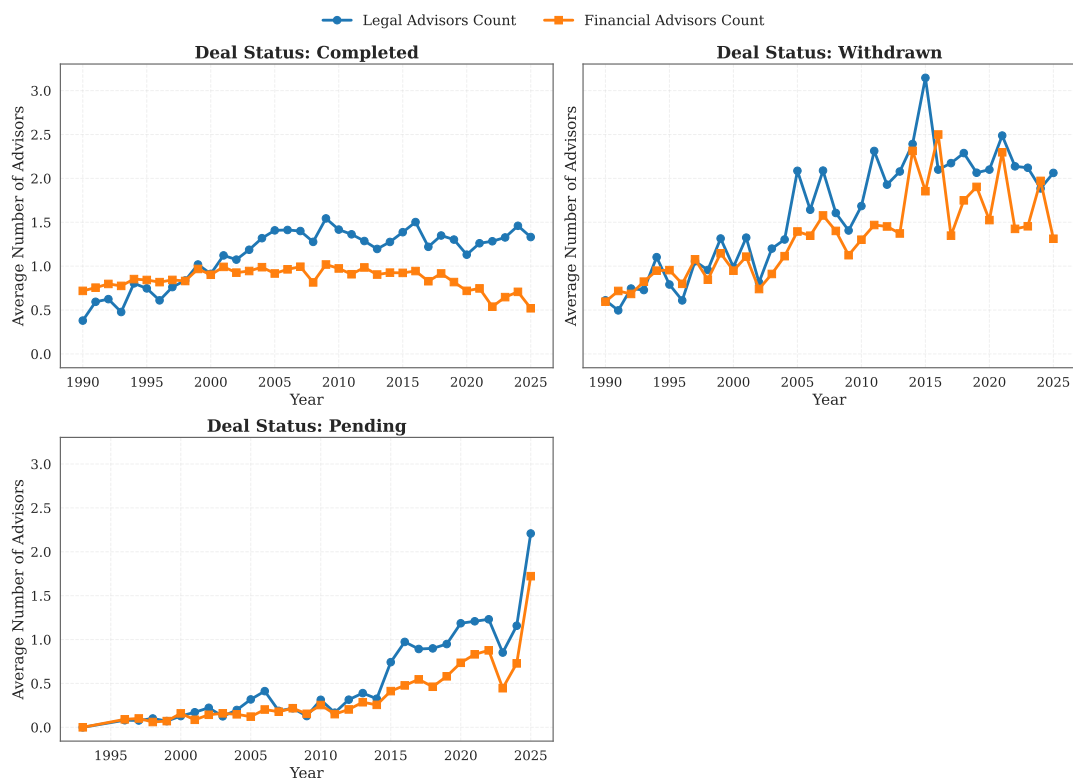


Figure 5.9 Average Number of Financial and Legal Advisors by Deal Status

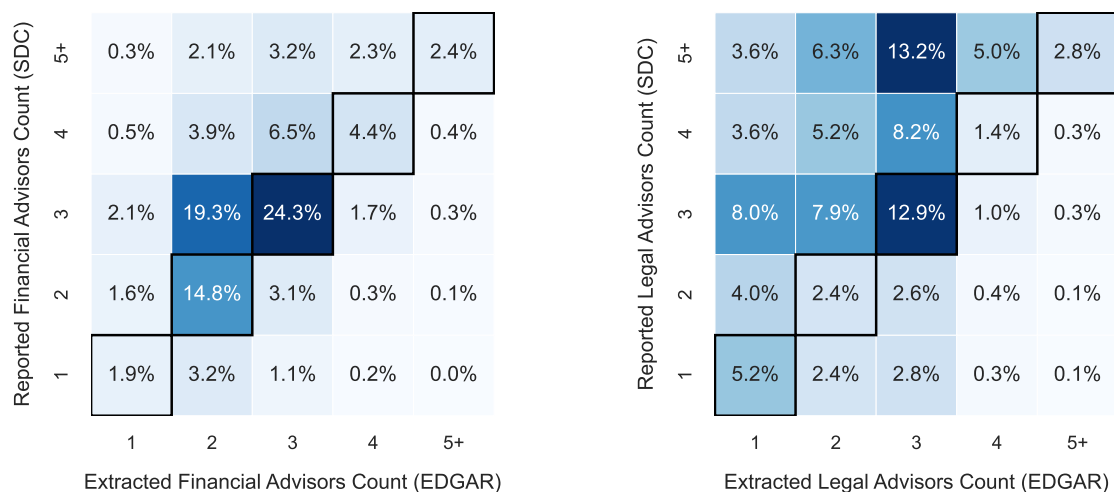
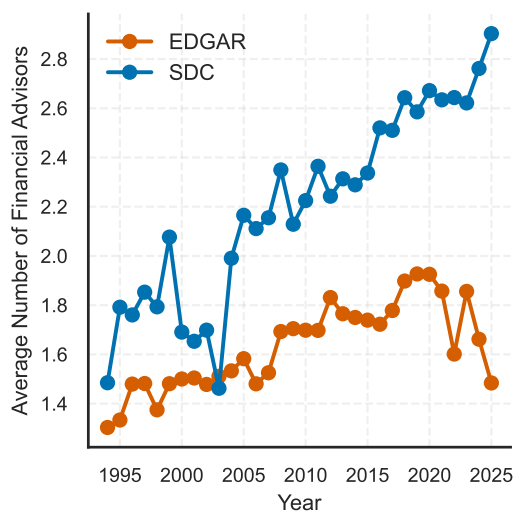
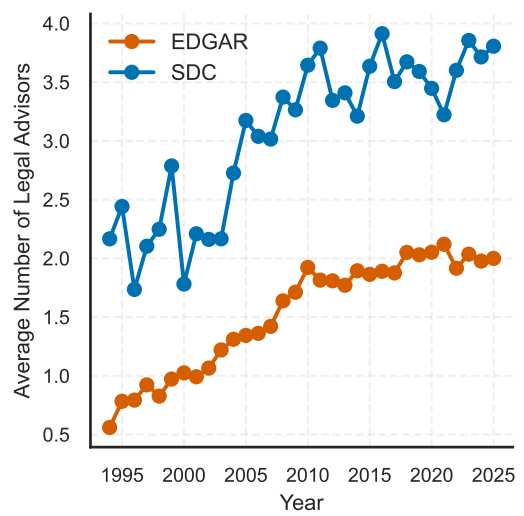


Figure 5.10 Comparison of financial and legal advisor coverage across EDGAR and SDC datasets



Panel A



Panel B

Figure 5.11 Comparison of advisor involvement per deal over time in the matched SDC-EDGAR sample. Panel A reports the average number of financial advisors per transaction by year, while Panel B reports the average number of legal advisors per transaction by year. Each panel also includes a fitted linear time trend.

5.4 From Events to Prices: Price-Specific Offers by Bidders

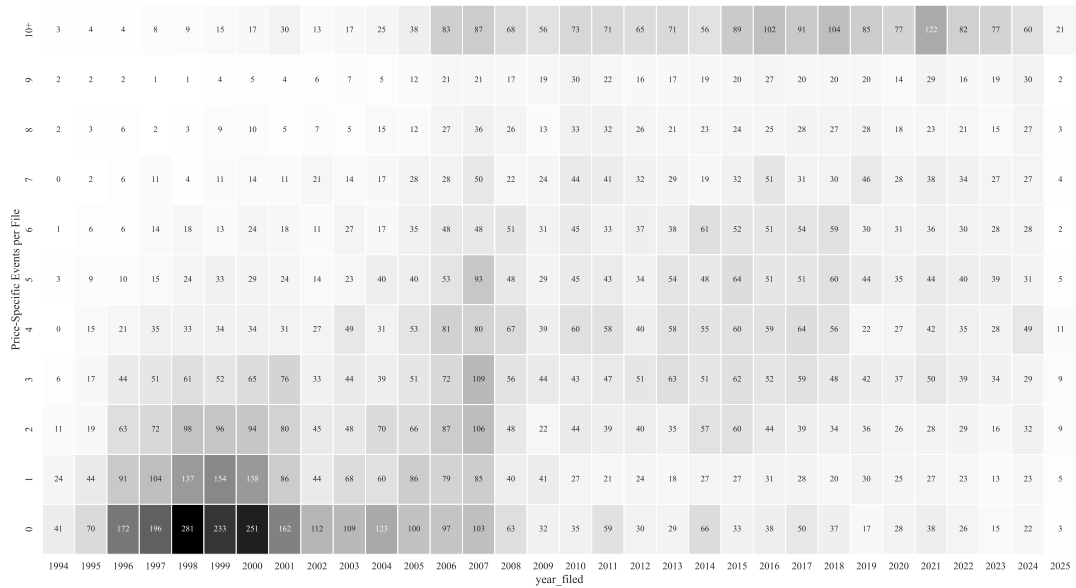
In this section, we focus on *Price-Specific Offers by Bidders* events in more detail. These events include dollar-denominated prices, including per-share offers as well as total transaction values expressed in millions or billions of dollars. We use this information to analyze rivalry among bidders and to examine how competition among bidders in the M&A market has evolved.

Figure 5.12 reports the number of price-specific offers by bidder events across years. Starting from 1994, the number of such events increased proportionally. The figure shows both the raw number of files per year (Panel A) and the normalized number across each year (Panel B).

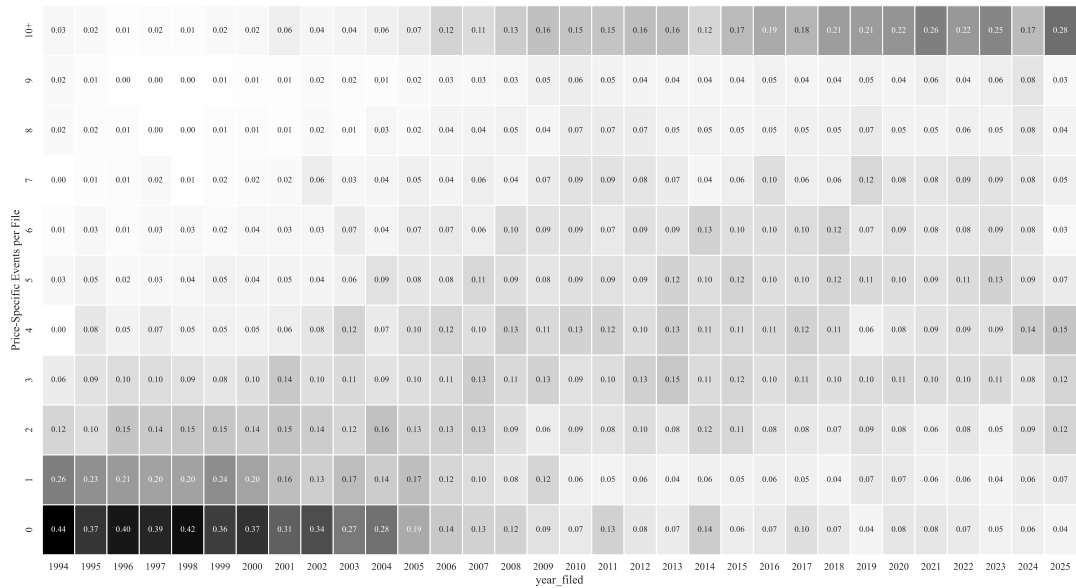
Figure 5.13 expands on the price analysis of price-specific offers by bidders' events. Panel A clearly shows that the number of mentioned prices in such events increased over the years with a little bit of a slowdown from 2008 to 2020. Panel B demonstrates that prices can be categorized as prices per share and total amounts in millions and billions of dollars. The deal-level dispersion of log prices indicates that, within some transactions, both per-share prices and total deal values are reported, implying mixed pricing units within the same deal.

Table 5.2 classifies deals into three categories: prices below \$500, prices above \$500, and mixed-price deals. As shown in Panel B, the majority of deals fall into the below-\$500 category, which also includes the 13.22% of deals for which no prices were reported. Panel C highlights that, within mixed-price deals, most individual price observations are below \$500. Furthermore, Panel A suggests that, as the number of price observations per deal increases, the share of mixed-price deals rises correspondingly, reflecting greater price heterogeneity in deals with higher price counts.

Consecutive revisions (Table 5.3) and revisions measured from the initial offer (Table 5.4) show that bidder competition is associated with larger and more dispersed price revisions, but the pattern differs sharply by price regime. For the "all-above-\$500" component, the mean revision and the upper-tail percentiles rise with the number of bidders, while medians remain much smaller than means, indicating strong right skew driven by occasional very large upward adjustments. The below-threshold components (both in all-below and mixed deals) medians cluster near 1-2%, interquartile ranges are tight, and standard deviations are substantially lower. A

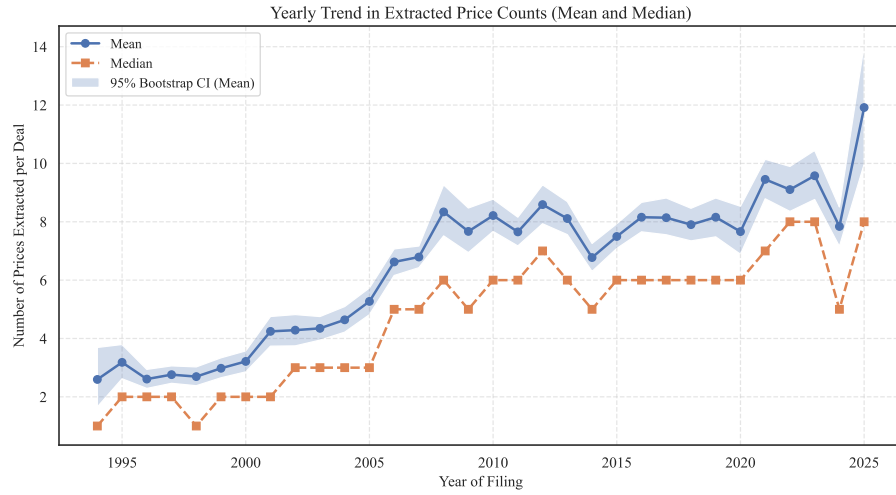


Panel A: Raw Counts

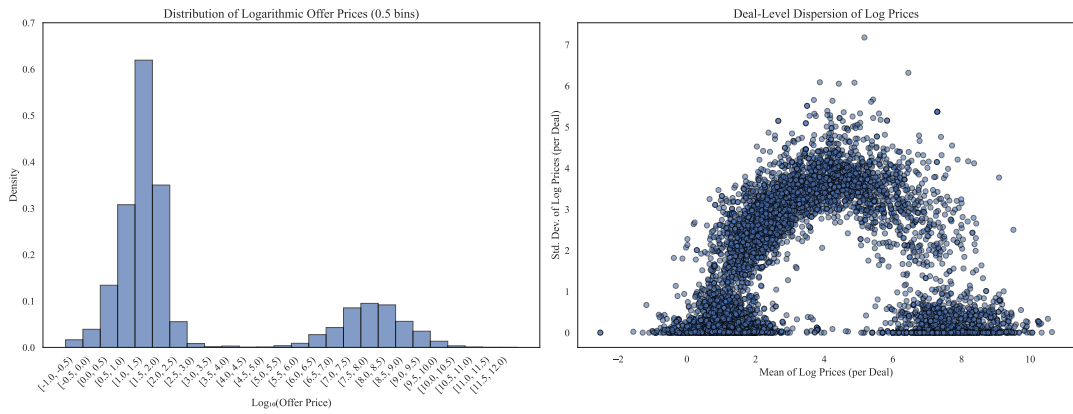


Panel B: Year-Normalized Proportions

Figure 5.12 Distribution of price-specific events across files. Panel A reports the raw count of price-specific events within each year-deal cell. Panel B reports the year-normalized distribution, where values represent within-deal proportions and allow comparison of relative timing patterns across transactions.



Panel A: Yearly Trend in Extracted Offer Prices



Panel B: Distribution and Dispersion of Offer Prices

Figure 5.13 Price offer extraction patterns. Panel A reports the yearly trend in the number of extracted prices per deal, where the solid line indicates the mean, the dashed line indicates the median, and the shaded area represents the 95% bootstrap confidence interval around the mean. Panel B reports the density distribution of log-transformed offer prices (left) and deal-level dispersion of log prices, showing the relationship between the mean and standard deviation of log prices within each deal (right).

Table 5.2 Offer Price Counts and Composition Across Deals

Panel A: Price Counts							Panel B: Deal Composition		
Count	Deals	%	<500	≥500	Mixed	Cum. %	Price Type	# Deals	Share (%)
0	1,559	13.22	100.0	0.0	0.0	13.22	All <500	7,446	63.13
1	1,191	10.10	70.11	29.89	0.0	23.31	All ≥500	1,139	9.66
2	1,062	9.00	69.02	20.43	10.55	32.32	Mixed	3,210	27.21
3	970	8.22	71.44	12.89	15.67	40.54	Panel C: Mixed Prices		
4	960	8.14	66.35	10.73	22.92	48.68	Category	# Obs	Share (%)
5	791	6.71	68.02	7.84	24.15	55.39	<500	24,667	64.54
6	718	6.09	59.33	9.89	30.78	61.48	≥500	13,551	35.46
7	615	5.21	57.72	6.50	35.77	66.69			
8	603	5.11	54.06	4.31	41.63	71.80			
9	479	4.06	51.77	3.76	44.47	75.86			
10	405	3.43	48.15	4.20	47.65	79.30			
11	387	3.28	48.84	3.62	47.55	82.58			
12	305	2.59	49.18	3.61	47.21	85.16			
13	245	2.08	43.27	4.49	52.24	87.24			
14	198	1.68	39.39	3.54	57.07	88.92			
15	162	1.37	35.80	6.17	58.02	90.29			
16	159	1.35	37.74	4.40	57.86	91.64			
17+	986	8.36	26.37	4.46	69.17	100.0			

Notes: Panel A reports the distribution of price-specific offers per deal. Panel B reports deal-level price-type classification. Panel C reports the distribution of individual price observations within mixed-price deals only.

striking complementary result is that the number of revisions per deal rises monotonically with bidder counts in every panel (Revs/Deal increases substantially for high-competition bins), implying that intensified competition both generates more bargaining rounds and increases the chance of extreme upward corrections in the above-threshold components. Comparing the two measures of revision (consecutive vs. from initial offer) confirms that cumulative revisions from the initial offer exaggerate the same asymmetries: means, upper percentiles, and standard deviations are generally larger when measured from the initial offer. Two caveats are important: sample sizes vary across bidder bins and regimes (the all-below panels contain far more observations than the all-above panels), and large standard deviations indicate outliers substantially influence means—therefore medians and percentiles provide a more robust summary of typical behavior, while the P75–P90 and Std highlight the increasing heterogeneity produced by competition.

Figure 5.14 presents revision-level renegotiation outcomes separately for below-threshold and above-threshold deals. Panel A reports deals with prices below the threshold, while Panel B reports deals with prices above the threshold. Below-threshold deals exhibit higher revision activity and greater variation in renegotiation outcomes, with both positive and negative revisions occurring throughout the sample period. In contrast, above-threshold deals display lower revision frequency

Table 5.3 Price Revision Statistics by Bidder Competition and Price Regime - Consecutive Price Revisions

Statistic	0	1	2	3	4	5	6+
<i>Panel A: All Prices Above Threshold</i>							
Mean	14.28	11.82	9.64	15.52	15.64	16.39	24.65
Median	2.00	0.00	1.06	5.21	6.90	9.31	7.28
P10	-37.59	-36.84	-38.99	-30.68	-35.34	-34.48	-38.46
P25	-15.90	-16.81	-19.90	-9.18	-11.20	-9.01	-16.56
P75	22.95	20.00	20.00	20.00	26.25	33.78	50.00
P90	90.06	67.18	59.05	73.86	82.88	72.91	108.14
Std	50.34	50.74	46.53	47.27	47.87	45.00	59.39
<i>N</i>	1,099	319	249	183	154	142	274
Revs/Deal	2.79	3.58	3.61	4.07	4.97	6.45	7.03
<i>Panel B: All Prices Below Threshold</i>							
Mean	6.50	5.35	4.15	3.64	3.25	2.96	2.69
Median	1.82	2.12	1.67	1.69	1.47	1.40	1.44
P10	-14.29	-14.29	-16.67	-15.42	-13.86	-16.43	-15.91
P25	-0.85	-2.17	-3.91	-3.92	-3.77	-4.35	-4.60
P75	8.33	8.70	8.33	7.62	7.41	7.50	7.14
P90	25.00	23.24	20.00	19.07	16.78	18.23	16.67
Std	27.86	24.47	25.10	22.24	20.39	21.22	19.56
<i>N</i>	9,109	3,879	3,701	2,229	1,928	1,456	4,344
Revs/Deal	3.84	4.82	6.43	6.44	8.14	8.37	9.67
<i>Panel C: Mixed Deals – Above Threshold</i>							
Mean	15.01	17.05	15.78	12.33	17.15	13.97	13.79
Median	2.10	3.70	2.32	3.68	2.04	1.76	2.92
P10	-35.71	-29.39	-30.11	-28.27	-34.90	-33.33	-32.68
P25	-10.00	-6.41	-7.43	-5.83	-10.49	-11.68	-13.37
P75	23.48	25.00	25.00	19.05	27.56	22.50	26.74
P90	90.00	84.57	78.40	66.67	100.00	74.91	73.18
Std	48.39	47.73	47.44	40.54	50.92	48.20	46.02
<i>N</i>	1,598	643	689	668	611	542	1,816
Revs/Deal	2.68	2.44	3.10	3.26	3.94	4.55	5.00
<i>Panel D: Mixed Deals – Below Threshold</i>							
Mean	6.01	7.90	4.80	4.32	4.85	4.35	4.15
Median	0.91	1.57	1.24	0.94	1.56	1.64	1.64
P10	-16.67	-20.00	-16.01	-16.67	-16.95	-20.00	-20.00
P25	-2.00	-3.74	-3.82	-3.75	-4.29	-6.25	-6.59
P75	8.15	10.23	8.33	8.33	9.09	9.79	9.52
P90	27.27	32.49	22.27	23.35	22.50	23.52	23.08
Std	28.50	33.59	25.75	24.31	25.51	26.29	26.49
<i>N</i>	3,737	2,220	2,198	1,911	1,718	1,709	4,933
Revs/Deal	4.31	5.21	7.00	6.64	8.03	9.88	10.11

Table 5.4 Price Revision Statistics by Bidder Competition and Price Regime - Revisions from Initial Offer

Statistic	0	1	2	3	4	5	6+
<i>Panel A: All Prices Above Threshold</i>							
Mean	8.77	11.30	17.35	9.52	22.58	28.58	29.77
Median	2.66	3.59	4.75	4.76	6.00	13.82	15.00
P10	-37.33	-26.99	-33.33	-37.72	-25.00	-35.97	-34.74
P25	-13.32	-9.09	-8.97	-11.73	-8.33	-9.81	-16.38
P75	20.00	24.58	27.27	19.52	50.00	50.00	66.67
P90	50.00	54.66	87.23	50.54	90.00	114.52	122.22
Std	40.63	36.26	49.84	42.08	47.78	55.92	60.04
N	1,076	346	265	187	165	150	246
Revs/Deal	2.74	3.76	3.96	4.07	5.32	6.82	6.83
<i>Panel B: All Prices Below Threshold</i>							
Mean	10.84	12.20	11.16	10.04	6.23	6.63	5.90
Median	6.67	7.24	6.67	6.06	4.40	6.67	4.17
P10	-13.36	-15.25	-18.26	-15.00	-20.70	-22.48	-20.75
P25	0.00	0.00	-0.63	0.00	-5.09	-5.77	-5.90
P75	17.02	19.25	17.95	16.67	16.67	18.32	14.29
P90	35.71	40.09	36.36	35.00	31.25	33.33	31.22
Std	27.47	29.55	30.65	27.09	23.52	23.36	24.81
N	9,129	3,877	3,695	2,264	1,936	1,428	4,356
Revs/Deal	3.86	4.82	6.45	6.64	8.20	8.35	9.70
<i>Panel C: Mixed Deals – Above Threshold</i>							
Mean	12.10	18.14	15.06	14.25	14.32	20.94	20.83
Median	4.17	5.80	5.33	6.38	0.00	13.72	5.56
P10	-36.96	-30.22	-30.00	-34.34	-36.00	-29.82	-34.80
P25	-10.71	-6.67	-9.42	-9.09	-18.18	-1.76	-16.67
P75	25.00	35.71	27.08	25.00	32.04	40.00	39.83
P90	66.67	68.32	68.75	68.00	90.48	75.15	100.00
Std	41.96	45.89	44.50	43.69	49.37	44.77	55.93
N	1,582	633	691	682	635	490	1,741
Revs/Deal	2.68	2.43	3.16	3.26	4.15	4.34	5.08
<i>Panel D: Mixed Deals – Below Threshold</i>							
Mean	10.08	14.27	10.91	14.42	14.64	9.17	7.75
Median	6.06	6.51	6.67	9.09	8.01	6.25	3.57
P10	-16.67	-24.62	-20.00	-15.40	-17.64	-27.27	-25.53
P25	0.00	-3.32	-0.73	0.00	-2.75	-5.84	-10.00
P75	17.10	22.09	20.00	25.00	25.99	21.09	17.50
P90	33.75	54.76	40.19	50.00	48.15	40.00	42.78
Std	28.95	40.06	29.91	31.07	35.23	31.28	31.84
N	3,751	2,241	2,189	1,799	1,702	1,632	4,685
Revs/Deal	4.40	5.37	7.04	6.71	8.14	9.77	9.99

and more limited renegotiation dynamics.

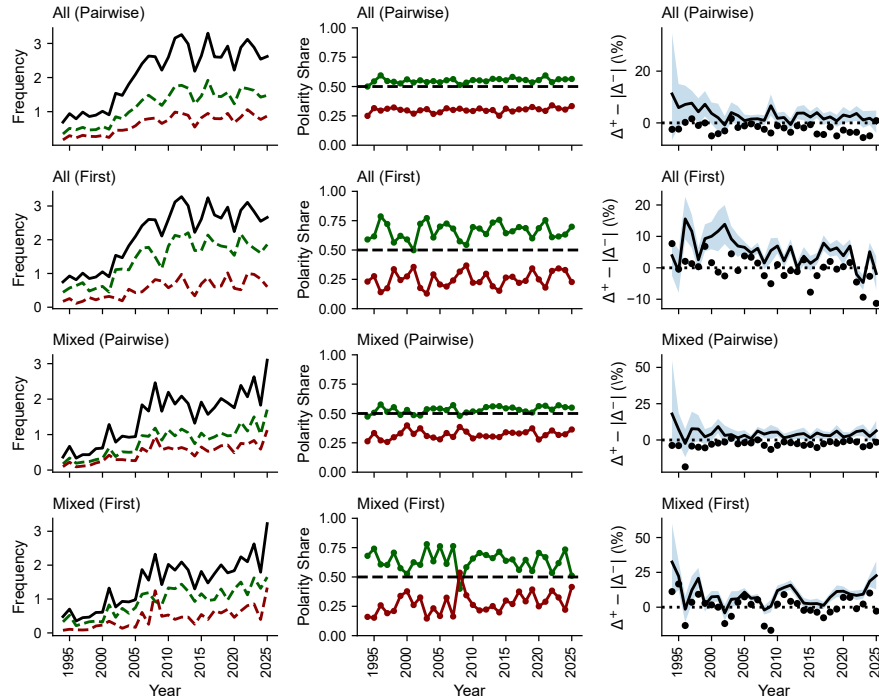
Across both regimes, the share of positive and negative revisions remains relatively balanced, although the magnitude of revisions differs across price regimes. Overall, the figure highlights systematic differences in renegotiation intensity and adjustment patterns between below-threshold and above-threshold deals.

Figure 5.15 illustrates the dynamics of percentage price revisions across sequential negotiation steps by deal price regime. Panels A1 and A2 present deals involving below-\$500 prices, including all-below-threshold deals and mixed-price deals with below-threshold components. Panels B1 and B2 present corresponding results for above-\$500 prices, including all-above-threshold deals and mixed-price deals with above-threshold components. Revisions relative to the initial offer exhibit larger cumulative price movements, whereas consecutive pairwise revisions are smaller in magnitude and more tightly distributed across revision steps. Pairwise revisions are generally symmetric between upward and downward adjustments, while revisions relative to the initial offer capture larger cumulative deviations from the starting price.

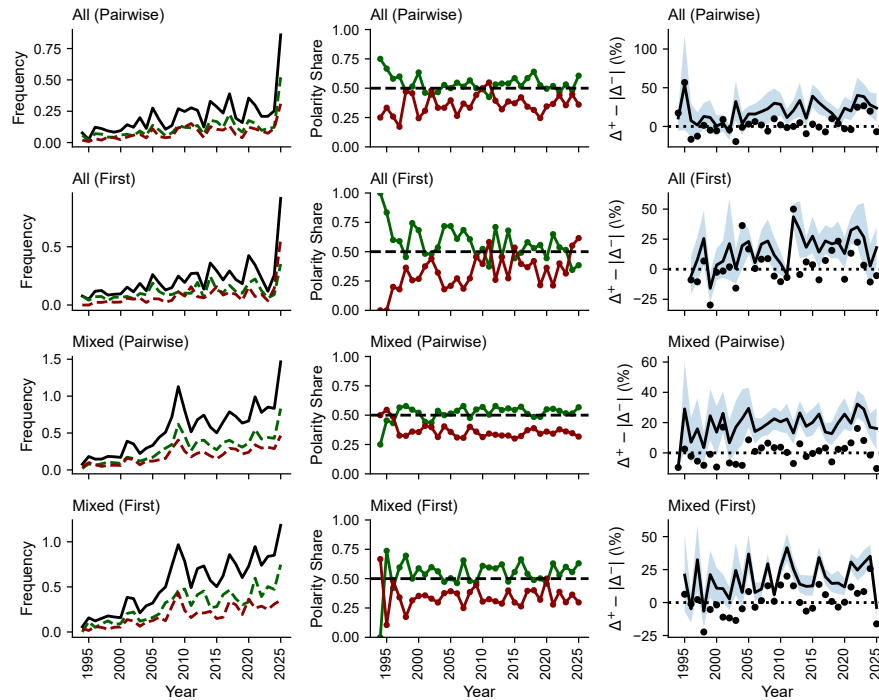
Table 5.5 summarizes the number of price revisions at each revision index, up to 20 revisions, for both pairwise and relative to first offer revisions. The majority of revisions occur within the initial rounds of the negotiation process, while observations become progressively fewer for higher-order revisions, particularly beyond the tenth revision index.

5.5 Regression Analysis

In this section, a regression analysis is performed to analyze whether extracted information using the LLM framework influences M&A takeover premiums. By incorporating financial control variables, we demonstrated that factors such as advisor involvement and deal timeline are drivers of premiums, providing insights for M&A analysts and shareholders.

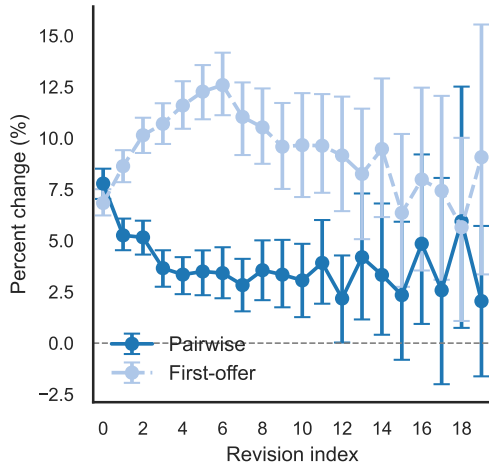


Panel A: Below-threshold deals

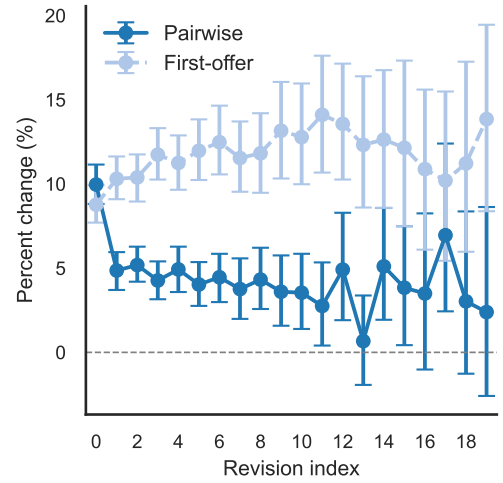


Panel B: Above-threshold deals

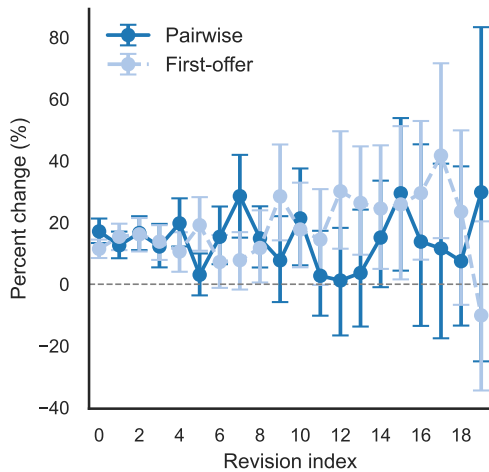
Figure 5.14 Revision-level price renegotiation dynamics by deal price regime. Panel A presents renegotiation outcomes for below-threshold deals, while Panel B presents outcomes for above-threshold deals. Each panel reports revision frequency, the share of positive and negative price revisions, and the magnitude differential between positive and negative revisions. Shaded areas denote bootstrap confidence intervals.



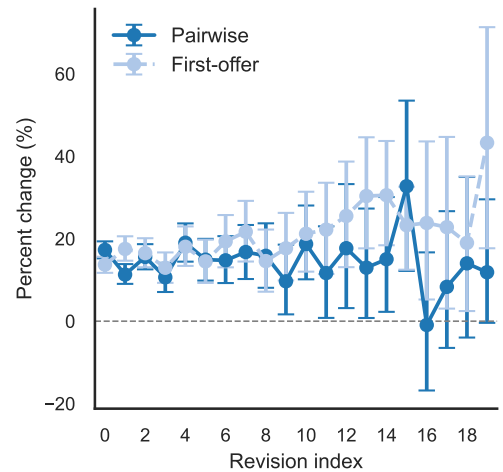
Panel A1: All Below



Panel A2: Mixed Below



Panel B1: All Above



Panel B2: Mixed Above

Figure 5.15 Percent changes in offer prices across sequential revision steps by deal price regime. Panels A1 and A2 report deals involving below-\$500 prices, while Panels B1 and B2 report deals involving above-\$500 prices. Pairwise revisions measure percentage changes between consecutive offers, whereas first-offer revisions measure percentage changes relative to the initial offer. Revision steps are aligned by event position and truncated at 20 revisions. Means are computed across deals at each revision index, with 95% bootstrap confidence intervals based on 1,000 resamples.

Table 5.5 Number of Observations by Revision Index

Revision	All Below		Mixed Below		All Above		Mixed Above	
	Pairwise	First	Pairwise	First	Pairwise	First	Pairwise	First
1	4957	4937	2771	2694	689	687	1925	1887
2	4165	4140	2406	2309	440	444	1213	1174
3	3466	3450	2103	2013	304	315	834	794
4	2777	2785	1788	1736	221	216	588	562
5	2222	2250	1523	1478	178	158	427	407
6	1807	1825	1279	1264	130	122	323	315
7	1475	1493	1101	1070	102	92	263	254
8	1176	1182	916	905	70	70	195	197
9	933	944	743	736	59	65	149	151
10	751	759	615	612	50	49	120	130
11	592	587	507	503	45	42	98	109
12	476	468	430	427	36	38	76	83
13	378	376	362	364	27	32	61	69
14	300	304	307	296	21	26	51	56
15	243	246	255	250	13	23	40	45
16	194	199	204	206	10	16	34	36
17	157	161	175	176	9	11	31	31
18	122	121	147	142	7	9	26	24
19	104	102	124	122	5	7	24	22
20	81	79	104	107	2	5	20	20

Table 5.6 OLS regression results for M&A takeover premiums.

Variables	1-Week Premium	4-Week Premium
Constant	67.762*** (5.05)	75.160*** (5.88)
Year Filed	-0.093 (0.10)	-0.285** (0.11)
Other Bidders Count	-0.505 (0.53)	-0.635 (0.59)
Financial Advisors Count	-3.632*** (1.06)	-4.697*** (1.22)
Legal Advisors Count	-0.372 (0.84)	-1.503 (0.93)
Price Specific Offers by Bidders	-0.120 (0.24)	-0.100 (0.28)
Confidentiality Agreement	1.337** (0.55)	1.349** (0.62)
Target Financial Advisor Hiring	-5.990** (2.36)	-4.556* (2.60)
Mean Log Lag	-2.414* (1.33)	-3.515** (1.53)
Tender Offer Flag	4.940** (2.34)	7.266*** (2.63)
Deal Value	-3.668*** (0.70)	-3.064*** (0.82)
Target ROA (1 Year Prior)	-0.274*** (0.05)	-0.349*** (0.06)
Acquirer Net Assets (1 Year Prior)	0.095 (0.35)	0.068 (0.41)
Observations	1974	1982
R^2	0.066	0.069
Adjusted R^2	0.060	0.063
F -Statistic	9.77***	10.48***

Notes: Heteroscedasticity-robust standard errors (HC3) are reported in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

6. DISCUSSION AND IMPLICATIONS

This chapter discusses and interprets the empirical results in economic terms. It positions the study's contribution to the literature, the importance of findings for empirical finance, limitations, and future research.

6.1 Interpretation of Deal Dynamics Findings

For parties involved in the private phase of mergers and acquisitions, the results show a skewed numerical distribution of other bidders and external financial and legal advisors, in which most of the other bidders' names are placeholders because of privacy issues, and for legal and financial advisors, some large corporations dominate the market. The temporal analysis of these entities shows a clear uptrend showcasing the increase in activity from external parties and making the private phase negotiations more active and intense.

Based on event definitions crucial for the private phase, we can observe an increase in the number of events with shorter inter-event gaps. The reason for this observation can be summarized as follows: first, the interplay of other bidders and external advisors with the target and final acquirer companies has increased the intensity of negotiations and the auction-like nature of deals. Second, with the advancement of technology, negotiation events have changed from formal in-person meetings to email, calls, and online meetings, increasing the number of events that fall into the definition of events.

The private phase of mergers and acquisitions typically progresses through a structured sequence of events. Target firms often retain financial advisors to manage

the sale process, after which potential bidders enter into confidentiality agreements to facilitate the exchange of private information. Following due diligence, bidders submit formal price-specific offers, either on a per-share basis or as total transaction values and share-based offers. Upon execution of a definitive agreement, the transaction is publicly announced. This sequence, *Target Financial Advisor Hiring → Confidentiality Agreement → Price-Specific Offers → Deal Announcement*, reflects an economically meaningful progression designed to reduce information asymmetry and enable price discovery. *Price-specific offers* mostly happen a few weeks before the deal announcement, showcasing the short time between offers and deal completion.

As the number of price-specific offers by bidders' events and the number of prices mentioned in such events have increased over time, it suggests that the private phase of M&A has become more competitive and supports the other bidders' increase over the years. The bimodality of price mentions in these events represents the per-share values and aggregate asset or enterprise values within the narratives.

The positive association between price revisions, time, and bidder competition is consistent with a dynamic bargaining interpretation of M&A negotiations. As the number of bidders increases, the mean and median price revisions for per-share prices generally decrease, consistent with internalizing the winner's curse and bid shading for preventing bidder loss. With the entry of one bidder into the private phase, the mean and median for per-share prices increase, showcasing the competition intensification between the final acquirer and the other bidder. The relatively larger positive revisions to negative revisions demonstrate an upward price revision tendency consistent with reducing information asymmetry during due diligence. Consistent with the increase in the number of other bidders over the years, the number of price revisions follows the same upward trend, suggesting a more competitive private phase of mergers and acquisitions. Consecutive price revisions show a more symmetric nature relative to the first price, indicating that the first price in deals acts more as an anchor for subsequent price revisions, but its effect fades away after a few revisions.

6.2 Contribution to the M&A Literature

This paper contributes to the M&A literature by providing large-scale evidence on private-phase negotiation dynamics based on SEC filings using large language models (LLMs). Traditionally, a strand of literature has begun to open the “black box” of the private phase of mergers and acquisitions by manually extracting information from SEC merger documents. A. L. Boone and Mulherin, 2007 documented that public takeover activities represent only a fraction of actual competition and distinguished between auctions and negotiations using hand-collected data. In another effort to test the winner’s curse hypothesis, the same authors A. L. Boone and Mulherin, 2008 used hand-collected data based on 308 deals. Aktas et al., 2009 showed that latent competition increases bid premiums, while Liu et al., 2023 documented frequent bid revisions during the private phase. Liu et al., 2022 claimed that takeover processes are fluid and introduced the concept of "negotiauction," and A. Boone et al., 2024 highlighted the strategic importance of initial bid strength in shaping subsequent bidding dynamics.

Despite these important advances, existing studies rely on labor-intensive hand collection and are therefore constrained to relatively small samples and human judgment and subjectivity. Moreover, prior work typically measures competition using summary indicators (e.g., number of bidders, confidentiality agreements signed), leaving the event-level sequence of negotiations, such as timing, bid revisions, and intermediate bargaining steps, largely unobserved in scalable datasets.

This study contributes by introducing a narrative-based multistage information extraction framework that systematically extracts structured negotiation events and different roles and entities from mergers and acquisitions SEC disclosures. By transforming unstructured narrative text into event-level measures of bidder contact, confidentiality agreements, private bids, price revisions, and process transitions, the study enables large-scale analysis of the microstructure of M&A bargaining. In doing so, it extends prior private-phase studies beyond manual coding and provides a scalable approach to studying negotiation dynamics across a broad universe of transactions and for a longer timespan.

More broadly, this approach allows the private phase of M&A transactions to be analyzed not merely as a binary auction-versus-negotiation classification but as a dynamic bargaining process observable through narrative disclosures.

6.3 Methodological Implications of LLM-Based Narrative Extraction

This study contributes to research methodology by demonstrating that large language models (LLMs) can systematically transform unstructured corporate disclosures into structured economic variables. Traditionally, variables related to negotiation dynamics, such as formal bids and confidentiality agreements, have required labor-intensive manual coding and, in some cases, such as event studies, were impossible. The LLM-based framework introduced here enables scalable and replicable extraction of such information directly from narrative disclosures.

A key methodological contribution is the implementation of a divide-and-conquer architecture. By chunking lengthy filings and structuring the extraction process into multiple stages, each focused on a specific type of information, the framework mitigates context-length limitations and increases inference accuracy.

The approach has several implications for empirical finance. First, it substantially lowers the cost of scaling qualitative research, making it feasible to construct large-sample datasets from textual disclosures. Second, it enhances reproducibility relative to hand-collected data, as extraction rules are encoded in prompts and model configurations rather than in subjective researcher interpretation. Third, while LLM-based measurement introduces model-driven error, such error is systematic and can be benchmarked against validation subsets.

6.4 Limitations and Directions for Future Research

This study faces several limitations. First, extracting background and introduction sections from SEC filings can be challenging due to variations in document structures, HTML formatting, and section boundaries, which may affect downstream LLM-based inference. Second, the context-length limitations of the Qwen3-8B model may reduce performance for very long documents, particularly when important information is distributed across distant sections. Third, LLM outputs may exhibit stochastic variation and potential hallucinations, especially in ambiguous legal or financial contexts, which can introduce noise into extracted variables. Fourth, the model is not specifically fine-tuned for M&A analysis and may not fully capture

domain-specific terminology or contractual nuances. Finally, extraction outcomes can be influenced by prompt design.

Future research on analyzing EDGAR filings can be summarized along three complementary directions. First, methodological expansion may involve the development of agentic Retrieval-Augmented Generation (RAG) systems in which multiple specialized agents iteratively retrieve, reason, and verify information from SEC filings. Such frameworks can autonomously identify economically relevant sections and generate evidence-grounded structured outputs for downstream financial analysis. Domain-adaptive training or fine-tuning of large language models on SEC-specific corpora, particularly merger and acquisition documents, may further enhance financial reasoning and reduce hallucination.

Second, computational scaling remains critical, as SEC filings frequently exceed standard context limits. Future research may explore long-context transformer architectures, hierarchical retrieval mechanisms, and memory-augmented systems to process long-form regulatory documents while preserving semantic coherence and inference reliability.

Third, economic validation should assess whether AI-extracted signals from EDGAR filings generate economically meaningful outcomes, such as improved prediction of bid premiums, abnormal returns, or deal completion probabilities.

6.5 Summary and Implications

This study provides large-scale evidence on the structure of private-phase negotiation dynamics in mergers and acquisitions and analyzes the interplay of entities by applying large language models to the narrative “Background of the Merger” disclosures contained in SEC filings. By systematically extracting structured information, including the target and final acquirer firms, other bidders, external advisors, negotiation events and their categorization, and price offers by bidder along with corresponding price revisions, the analysis demonstrates that these disclosures encode economically meaningful information about the microstructure of the private phase of mergers and acquisitions. The results show that the private phase of M&As can be analyzed at scale using artificial intelligence within a structured LLM-based extraction framework.

The primary research contribution of this study lies in expanding the empirical frontier of private-phase mergers and acquisitions analysis. Traditionally, studies examining negotiation dynamics have relied on labor-intensive manual analysis of a limited number of transactions, restricting both sample size and the ability to study dynamic patterns across time within transactions and across transactions. By contrast, the LLM-based framework introduced here enables scalable and replicable extraction of negotiation events from thousands of filings.

Methodologically, this study demonstrates how large language models can transform narrative disclosures into structured datasets suitable for econometric analysis. It shows that narrative corporate filings can be mined for verifiable institutional events and sequential negotiation details. The contribution therefore extends beyond the specific application: **it illustrates a general framework for converting qualitative disclosures into structured economic variables, thereby reducing reliance on manual coding.**

The implications extend to future research on mergers and acquisitions and corporate finance more broadly. With scalable access to negotiation-level variables, researchers can examine how bargaining structure interacts with governance, information asymmetry, industry competition, macroeconomic conditions, and post-merger performance. Moreover, the integration of narrative-based microstructure variables with traditional deal-level data enables richer identification strategies and more precise modeling of transaction dynamics.

The findings suggest that advances in language modeling expand the scope of empirical analysis into the private phase of mergers and acquisitions. By making negotiation processes observable and measurable at scale, LLM-based extraction techniques change the focus of M&A research from static transaction features toward dynamic, process-based analysis.

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APPENDIX A - PROMPTS USED FOR THE MULTI-STAGE LLM-BASED FRAMEWORK

Target and Final Acquirer Extraction Prompt

Objective. Identify the target and the final acquirer firms associated with the transaction described in the SEC filing, using only the introductory portion of the document.

Prompt Template.

NOTE: This section contains only the introductory part of the filing, including company information and a letter to stockholders regarding the deal, provided before the table of contents. Use only this portion to answer.

Your task: Read the text carefully and identify the **target** firm and the **final acquirer**, based only on the evidence in the text.

Return your result in JSON format matching the schema below.

JSON Schema:

```
{
  "target": {
    "description": "Full, official name of the company that is
    ↪ identified or approached for acquisition or merger by
    ↪ another firm. It is the entity whose ownership interest
    ↪ (stock or assets) is sought to be purchased, merged, or
    ↪ controlled."
  },
  "final_acquirer": {
    "description": "Full, official name of the company that
    ↪ seeks to execute and ultimately completes the
    ↪ acquisition or merger by making an offer to purchase,
    ↪ merge with, or combine with the target firm."
  }
}
```

Expected Output Example:

```
{
  "target": "Widget Co.",
  "final_acquirer": "Omni Holdings"
}
```

Definitions:

- **Target firm:** The company whose ownership interest (stock or assets) is purchased, merged, or controlled by another firm. Its shareholders typically receive shares of the acquirer or cash in a merger or acquisition.
- **Final acquirer:** The ultimate parent company that seeks to complete the acquisition or merger.

Important: If the deal uses a shell, subsidiary, or merger vehicle (e.g., “Merger Sub,” “Acquisition Corp.” created only for this transaction), do *NOT* list that entity as the final acquirer or target. Instead, identify the parent company controlling it.

Rules:

1. Ignore committees, departments, sub-units, and financial or legal advisors (e.g., banks, law firms).
2. If a shell company, acquisition vehicle, or wholly owned subsidiary is mentioned, treat it as a transaction vehicle only. The real final acquirer is its parent company.
3. Use exact, full legal names as written in the text. Do not abbreviate or alter them.
4. Respond only with valid JSON. Do not include explanations.

Text to analyze:

[Introductory section text inserted here]

Expected Output Format.

```
{
  "target": "...",
  "final_acquirer": "..."
}
```

Other Bidders and External Advisor Extraction Prompt

Objective. Jointly extract the names of other bidding parties and the external financial and legal advisors involved in the transaction.

Prompt Template.

You are an information-extraction assistant.

- Use the **Introductory Information** only to determine the Target and Final Acquirer.
- Use the **Background of Deal** only to extract other bidders, financial advisors, and legal advisors.

Do *not* invent facts or add placeholders unless the document explicitly uses them.

Definitions:

- **Other Bidders:** Companies that attempt to acquire a target but fail, often due to higher competing bids, regulatory hurdles, or strategic misalignments. These include companies or entities that submitted competing offers or were in discussions to acquire or merge with the target. Excludes the target, the final acquirer, their subsidiaries, shells, SPVs, internal committees, counsel groups, and any advisors. Only organization names or placeholders; exclude person names.
- **Financial Advisors:** External financial institutions, banks, or advisory firms retained to provide valuation, deal structuring, and negotiation expertise. Excludes in-house teams, the target, the final acquirer, their subsidiaries, shells, SPVs, other bidders, and legal advisors. Only organization names; exclude person names.
- **Legal Advisors:** External law firms or legal counsel retained to manage legal due diligence, contract negotiations, and regulatory compliance. Excludes in-house legal departments, the target, the final acquirer, their subsidiaries, shells, SPVs, other bidders, and financial advisors. Only organization names; exclude person names.
- **Target firm:** The company whose ownership interest (stock or assets) is purchased, merged, or controlled by another firm.
- **Final acquirer:** The company that executes the acquisition or merger by purchasing, merging with, or combining with the target firm.

- **Shell company:** A company with little or no operations that serves as a legal or financial vehicle (e.g., SPAC, holding company, offshore shell). Do not confuse a shell with the true Target or Final Acquirer. If the shell is directly tied to the Target or Final Acquirer, exclude it.

— **INPUT** —

Introductory Information (for identifying Target and Final Acquirer):

[Introductory section text]

Target (already determined from the Intro section; exclude from all other roles, including subsidiaries, shells, and SPVs):

[Target firm name]

Final Acquirer (already determined from the Intro section; exclude from all other roles, including subsidiaries, shells, and SPVs):

[Final acquirer name]

Background of Deal (for extracting other bidders and advisors):

[Background-of-the-merger text]

— **END INPUT** —

OBJECTIVE (strict):

Return exactly one JSON object that conforms to the following JSON Schema:

```
{
  "other_bidders": [ "<bidder_name_1>", "<bidder_name_2>", ...
    ↪ ],
  "financial_advisors": [ "<advisor_name_1>",
    ↪ "<advisor_name_2>", ... ],
  "legal_advisors": [ "<advisor_name_1>", "<advisor_name_2>",
    ↪ ... ]
}
```

KEY RULES (enforce these precisely):

1. **Target and Final Acquirer**

- Already identified from the Intro section.
- Never include them, or any of their subsidiaries, shells, SPVs, or affiliates, in `other_bidders`, `financial_advisors`, or `legal_advisors`.

2. **Financial and Legal Advisors**

- Must be the official names as written in the document.

- Only external firms; exclude in-house teams, individuals, the Target, the Final Acquirer, or their subsidiaries, shells, or SPVs.

3. Other Bidders

- Include only organizations explicitly named as having submitted offers, made bids, expressed interest, or entered negotiations.
- Include placeholders only if they appear verbatim in the Background section (e.g., “Company A–Z,” “Third Party,” “Party A”).
- Do not invent placeholders.

4. Exclusions and Disambiguation

- Exclude internal committees, in-house counsel, individuals, and any entities already assigned to another role.
- Prefer omission over guessing if ambiguous.

5. Formatting

- Use only organization names or verbatim placeholders.
- De-duplicate entries and preserve order of first appearance.

EXAMPLES (valid outputs):

- If no bidders or advisors identified:

```
{ "other_bidders": [], "financial_advisors": [],  
  ↪  "legal_advisors": [] }
```
- If placeholders used:

```
{  
  "other_bidders": ["Company A", "Company B"],  
  "financial_advisors": ["Goldman Sachs, Inc."],  
  "legal_advisors": ["Latham & Watkins LLP"]  
}
```

OUTPUT FORMAT (MANDATORY):

- Return exactly one JSON object valid against the JSON Schema.
- No commentary, markdown, or text outside the JSON object.

Expected Output Format.

```
{  
  "other_bidders": [...],  
  "financial_advisors": [...],  
  "legal_advisors": [...]  
}
```

Private-Phase Event Extraction Prompt

Objective. Extract all private-phase interactions and negotiation events from the background-of-the-merger text and represent them as structured chronological entries.

Prompt Template.

Brief task description:

Given the provided transaction background text chunk, extract every interaction or event in the order it appears. For each event, produce one JSON object conforming to the ChronoEntry schema below.

JSON schema definition:

```
{
  "date": {
    "description": "Date or temporal cue for the event. Follow this
    ↳ precedence when extracting: (1) exact date (YYYY-MM-DD); (2)
    ↳ year and month (YYYY-MM); (3) exact date range, e.g., [From
    ↳ 2007-06-19 to 2007-06-21]; (4) continuous or period-based
    ↳ expressions, e.g., [During Q3 2004], [Throughout 2005]; (5)
    ↳ approximate or partial expressions, e.g., [Late 2003],
    ↳ [Around mid-July 2007]; (6) relative or sequential
    ↳ expressions, e.g., [After 2004-03-31]; and (7) indefinite or
    ↳ unspecified cases, using [Date unspecified]. Always prefer
    ↳ the most specific representation available."
  },
  "event": {
    "description": "A concise but complete single sentence describing
    ↳ the interaction, derived only from the provided background
    ↳ text. Include parties, the event type, and any concrete
    ↳ outcomes or financial terms. Do not invent facts or imply
    ↳ outcomes not present in the text; indicate missing
    ↳ information explicitly in brackets."
  }
}
```

}

HARD CONSTRAINTS (do not violate):

1) Output format

- Output **solely** a JSON array of objects (no extra text, markdown, commentary, or explanation).
- The array must contain one object per extracted event and nothing else.

2) Date rules (precedence and formatting)

- Prefer the most specific date available in the following order: exact (YYYY-MM-DD) → year-month (YYYY-MM) → ranges or quarters → approximate → relative → [Date unspecified].
- Always use brackets around non-exact or non-YYYY-MM-DD values (e.g., [Late 2003], [From 2005 to 2007], [After 2004-03-31]).
- For ranges, prefer the form [From YYYY-MM-DD to YYYY-MM-DD]. For non-standard temporal phrases, include the verbatim phrase in brackets.

3) Event definitions (what to extract)

- **Engagements:** Any meeting, call, site visit, or other direct engagement between two or more parties.
- **Agreements:** Any contract or agreement executed or signed (e.g., confidentiality agreements, formal bids, purchase agreements, acceptance letters).
- **Expressions of intent:** Any indication of interest to buy, sell, merge, acquire, divest, or otherwise combine businesses.
- **Financial terms:** Include verbatim any prices, share terms, consideration amounts, fee structures, or formulas.

4) Description content rules

- The description must be a single clear sentence derived only from the text chunk. Include parties (with roles when available), the nature of the event, and outcomes or key points.
- Do not invent facts or attributes not present in the chunk. If information is missing, mark it explicitly in brackets (e.g., [buyer name unknown]).
- If the chunk contains multiple small, tightly related actions that form one coherent episode (micro-events), include all such actions in the description while preserving sequence and outcomes.

5) Chronology and ordering

- Process the text chunk in the order sentences appear and output events in the same order.
- Do not reorder events.

6) Grouping and merging micro-events

- Merge micro-events into a single episode when they (a) occur within the same paragraph or within two consecutive sentences, (b) involve the same primary parties, and (c) represent steps of a single interaction (e.g., phone call → follow-up email → site visit).
- When merging, include all sentences or micro-events from the chunk in the **description**. You may concatenate them into one fluent sentence if necessary, but preserve the original sequence and explicit outcomes or terms.
- If micro-events are distinct in parties or outcomes, keep them separate.

7) Context and omission

- Do not omit events that satisfy the event definitions. Omission is permitted only for sentences that are purely narrative and contain no qualifying event.
- If an event lacks a required field (e.g., no date), still create an entry and use an appropriate bracketed placeholder (e.g., "date": "[Date unspecified]").

Single consolidated examples block (canonical format and grouping)

```
[
  {
    "date": "2004-12-16",
    "description": "Buyer submitted a written offer to Target's board
    ↪ proposing $25 per share in cash."
  },
  {
    "date": "[From 200X-06-01 to 200X-06-03]",
    "description": "On 200X-06-01 X called Y; on 200X-06-02 Y sent a
    ↪ draft NDA; on 200X-06-03 the parties agreed on terms."
  },
  {
    "date": "[Date unspecified]",
    "description": "Senior management of Target held exploratory
    ↪ discussions with unnamed Buyer representatives concerning a
    ↪ possible acquisition; no terms were finalized."
  }
]
```

Execution

Process the following chunk now. Output only a JSON array matching the ChronoEntry schema above; do not include any other text.

{chunk}

Event Categorization Prompt

Objective. Assign each previously extracted chronological event to one of five standardized private-phase event categories.

Prompt Template.

Brief Task Description:

For each chronological entry (with date and description) from a transaction's back-

ground documentation, assign it to exactly one of the five predefined event categories.

JSON Schema:

```
{
  "date": {
    "description": "Return the date field exactly as provided-do not
    ↪ modify its content."
  },
  "description": {
    "description": "Return the description field exactly as
    ↪ provided-do not modify its content."
  },
  "category": {
    "enum": [
      "Target Financial Advisor Hiring",
      "Confidentiality Agreement",
      "Price Specific Offers by Bidders",
      "Deal Announcement",
      "Other"
    ],
    "description": "One of the five predefined event categories."
  }
}
```

Illustrative Examples (drawn from various different deal files):

These are meant to show the correct way to assign categories to individual event entries. Each example stands on its own and is not part of the same transaction.

The following examples are drawn from different transactions and are provided solely to illustrate correct categorization. Each example is independent and does not belong to the same deal.

```
[
  {
    "date": "May 8, 1997",
```

```

    "description": "In light of continuing consolidation in
    ↪ the office products industry, Viking retained Merrill
    ↪ Lynch and Warburg Dillon Read on May 8, 1997 to
    ↪ explore strategic alternatives.",
    "category": "Target Financial Advisor Hiring"
  },
  {
    "date": "Early fall of 1998",
    "description": "In the early fall of 1998, Gulfstream
    ↪ retained Merrill Lynch & Co. and Goldman Sachs & Co.
    ↪ to assist Gulfstream in exploring strategic
    ↪ alternatives.",
    "category": "Target Financial Advisor Hiring"
  },
  {
    "date": "November 8, 2000",
    "description": "On November 8, 2000, Quaker and Danone
    ↪ entered into a confidentiality agreement.",
    "category": "Confidentiality Agreement"
  },
  {
    "date": "September 11, 2001",
    "description": "Mohawk and Dal-Tile executed a
    ↪ confidentiality agreement on September 11, 2001.",
    "category": "Confidentiality Agreement"
  },
  {
    "date": "Not specified",
    "description": "Following this meeting, Cardinal Health
    ↪ and Syncor executed a mutual confidentiality
    ↪ agreement in anticipation of further discussions to
    ↪ pursue the matters raised at the first meeting.",
    "category": "Confidentiality Agreement"
  },
  {
    "date": "Not specified",

```

```

        "description": "Cardinal Health offered that possible
        → exchange ratios of Syncor common stock for Cardinal
        → Health common shares in a potential combination in
        → the range of 0.57 to 0.62 subject to, among other
        → things, satisfactory completion of due diligence,
        → negotiation of definitive documents and related
        → matters.",
        "category": "Price Specific Offers by Bidders"
    },
    {
        "date": "September 29, 2006",
        "description": "On September 29, 2006, Angio sent a
        → proposal to RITA, which provided for a per share
        → purchase price of $3.90 per RITA share in stock and
        → cash.",
        "category": "Price Specific Offers by Bidders"
    },
    {
        "date": "September 7, 2004",
        "description": "Following approvals of the merger and
        → related transactions by the respective boards of
        → directors of Nanogen and Epoch, representatives of
        → Nanogen and Epoch finalized and executed the
        → definitive merger agreement in the afternoon of
        → September 7, 2004 and then issued a press release
        → announcing the execution of the merger agreement
        → after the close of the stock markets on September 7,
        → 2004.",
        "category": "Deal Announcement"
    }
]

```

Category Definitions:

1. **Target Financial Advisor Hiring:** Events that describe target firms' hiring of financial advisors or investment banks to explore or evaluate strategic alternatives, including mergers, acquisitions, or takeovers. Target firms may also hire financial advisors or investment banks after receiving an unsolicited offer.
2. **Confidentiality Agreement:** Execution of confidentiality agreements or

nondisclosure agreements (NDAs) between the target and the bidders. Captures both mutual and unilateral confidentiality undertakings.

3. **Price Specific Offers by Bidders:** Any proposal or expression of interest made by bidders that explicitly states a price, valuation, exchange ratio, or share-term. Includes letters of intent (LOIs) with stated price, term sheets quoting per-share amounts, or valuation ranges tied to consideration. Exclude counter offers or offer revision requests made by target firms.

4. **Deal Announcement:** Press releases, or regulatory filings (e.g., SEC) that publicly announce or confirm the execution of a definitive agreement between the target and the buyer firm.

5. **Other:** Relevant events not captured by the above definitions.

Process Guidelines

Before you assign any category, think through each event entry in chronological order:

1. Identify the key action.
2. item Match it to the definition of each of the five categories.
 - For **Target Financial Advisor Hiring**, categorize the event if **{Target}** hires any of **{Financial Advisors}**.
 - For **Price Specific Offers by Bidders**, categorize the event if **{Other Bidders}** or the **{Final Acquirer}** submitted any price-based or share-based offers to the **{Target}**.
 - *Do not count any offers that merely express interest or involve ongoing negotiations.*
3. Confirm you have exactly one best fit.
4. Return the **date** and **description** fields exactly as provided; do not modify their content.

Return a JSON array of objects with the fields **date**, **description**, and **category**. Use exactly one of the five labels above. If an event matches none of these definitions, assign **Other**.

Text (multiple events for this chunk):

[Events extracted from the filing, including dates and event
↔ descriptions, are inserted here.]

APPENDIX B - YEARLY TRENDS IN DEAL COMPETITION AND ADVISORY PARTICIPATION

Table B.1 Yearly summary statistics for the number of other bidders per deal

Year	Obs.	Mean	Median	Std.	Min	P25	P75	Max	IQR	Range	Skew	Kurt.
1994	94	0.22	0.0	0.49	0	0.0	0.0	2	0.0	2	2.15	3.98
1995	192	0.29	0.0	0.86	0	0.0	0.0	5	0.0	5	3.85	15.92
1996	428	0.32	0.0	0.93	0	0.0	0.0	8	0.0	8	4.48	25.42
1997	516	0.42	0.0	0.93	0	0.0	0.0	6	0.0	6	3.07	11.34
1998	677	0.34	0.0	0.79	0	0.0	0.0	6	0.0	6	3.13	11.68
1999	660	0.34	0.0	0.89	0	0.0	0.0	7	0.0	7	3.80	18.91
2000	685	0.36	0.0	0.92	0	0.0	0.0	8	0.0	8	3.59	16.69
2001	538	0.48	0.0	1.18	0	0.0	0.0	10	0.0	10	4.08	22.90
2002	338	0.50	0.0	1.06	0	0.0	1.0	8	1.0	8	2.98	11.57
2003	416	0.58	0.0	1.27	0	0.0	1.0	10	1.0	10	3.70	18.85
2004	451	0.60	0.0	1.25	0	0.0	1.0	7	1.0	7	2.59	6.74
2005	527	0.71	0.0	1.44	0	0.0	1.0	11	1.0	11	3.34	14.66
2006	688	1.22	0.0	2.15	0	0.0	2.0	17	2.0	17	2.81	11.25
2007	837	1.35	0.0	2.48	0	0.0	2.0	26	2.0	26	3.84	23.49
2008	517	1.46	0.0	2.11	0	0.0	3.0	12	3.0	12	1.61	2.49
2009	355	1.58	1.0	2.44	0	0.0	2.0	17	2.0	17	2.34	6.93
2010	485	1.78	1.0	2.67	0	0.0	2.0	15	2.0	15	2.02	4.36
2011	476	1.70	0.0	2.73	0	0.0	2.0	15	2.0	15	2.32	5.96
2012	400	1.87	0.0	2.99	0	0.0	3.0	17	3.0	17	2.46	7.27
2013	449	2.47	1.0	3.81	0	0.0	3.0	23	3.0	23	2.51	7.70
2014	491	1.30	0.0	2.14	0	0.0	2.0	13	2.0	13	2.24	5.90
2015	531	2.11	1.0	3.14	0	0.0	3.0	17	3.0	17	2.16	5.29
2016	545	2.47	1.0	3.75	0	0.0	3.0	27	3.0	27	2.72	9.87
2017	532	2.90	1.0	4.23	0	0.0	4.0	29	4.0	29	2.22	6.73
2018	515	2.43	1.0	2.96	0	0.0	4.0	18	4.0	18	1.61	2.99
2019	419	2.80	1.0	4.13	0	0.0	4.0	29	4.0	29	2.51	8.70
2020	355	2.14	1.0	3.15	0	0.0	3.0	28	3.0	28	2.86	14.09
2021	488	2.58	1.0	3.36	0	0.0	4.0	24	4.0	24	2.09	7.18
2022	382	2.54	1.0	3.29	0	0.0	4.0	19	4.0	19	1.83	3.82
2023	319	3.45	2.0	4.53	0	0.0	5.0	24	5.0	24	1.82	3.53
2024	367	3.15	2.0	4.99	0	0.0	5.0	52	5.0	52	4.36	30.91
2025	77	4.23	3.0	5.29	0	0.0	6.0	26	6.0	26	1.91	3.98

Table B.2 Yearly Distribution of Financial Advisors per Deal

Year	<i>N</i>	Mean	Median	Std.	Min	P25	P75	Max	IQR	Range	Skew	Kurt.
1994	94	1.23	1.0	1.01	0	1.0	2.0	5	1.0	5	0.98	1.34
1995	192	1.31	1.0	0.90	0	1.0	2.0	4	1.0	4	0.73	0.51
1996	428	1.39	1.0	0.87	0	1.0	2.0	4	1.0	4	0.29	-0.46
1997	516	1.39	1.0	0.89	0	1.0	2.0	7	1.0	7	1.01	3.40
1998	677	1.33	1.0	0.82	0	1.0	2.0	5	1.0	5	0.36	0.42
1999	660	1.43	1.0	0.84	0	1.0	2.0	5	1.0	5	0.63	1.13
2000	685	1.50	1.0	0.88	0	1.0	2.0	6	1.0	6	0.95	1.71
2001	538	1.40	1.0	0.82	0	1.0	2.0	6	1.0	6	0.64	1.81
2002	338	1.42	1.0	1.01	0	1.0	2.0	6	1.0	6	1.36	3.49
2003	416	1.31	1.0	0.81	0	1.0	2.0	5	1.0	5	0.75	1.15
2004	451	1.44	1.0	0.84	0	1.0	2.0	5	1.0	5	0.54	0.63
2005	527	1.50	1.0	0.87	0	1.0	2.0	5	1.0	5	0.83	1.55
2006	688	1.43	1.0	0.82	0	1.0	2.0	5	1.0	5	0.95	1.79
2007	837	1.53	1.0	0.91	0	1.0	2.0	8	1.0	8	1.46	6.41
2008	517	1.56	1.0	0.98	0	1.0	2.0	6	1.0	6	0.80	1.14
2009	355	1.66	2.0	0.94	0	1.0	2.0	5	1.0	5	0.67	0.76
2010	485	1.59	1.0	0.94	0	1.0	2.0	7	1.0	7	1.21	3.29
2011	476	1.57	1.0	0.93	0	1.0	2.0	6	1.0	6	0.85	1.89
2012	400	1.56	1.0	0.92	0	1.0	2.0	6	1.0	6	1.03	2.38
2013	449	1.76	2.0	1.05	0	1.0	2.0	7	1.0	7	1.26	2.91
2014	491	1.72	2.0	1.10	0	1.0	2.0	10	1.0	10	1.40	6.70
2015	531	1.73	2.0	0.94	0	1.0	2.0	6	1.0	6	0.88	1.75
2016	545	1.74	2.0	1.02	0	1.0	2.0	7	1.0	7	1.16	2.62
2017	532	1.72	2.0	0.96	0	1.0	2.0	6	1.0	6	0.95	2.11
2018	515	1.84	2.0	1.12	0	1.0	2.0	9	1.0	9	1.43	4.38
2019	419	1.83	2.0	1.07	0	1.0	2.0	7	1.0	7	1.14	2.58
2020	355	1.82	2.0	0.97	0	1.0	2.0	5	1.0	5	0.57	0.42
2021	488	1.78	2.0	1.01	0	1.0	2.0	6	1.0	6	0.81	1.15
2022	382	1.57	1.0	0.89	0	1.0	2.0	6	1.0	6	0.74	1.63
2023	319	1.78	2.0	1.07	0	1.0	2.0	9	1.0	9	1.58	7.11
2024	367	1.67	2.0	0.90	0	1.0	2.0	6	1.0	6	1.20	3.10
2025	77	1.62	1.0	1.03	1	1.0	2.0	6	1.0	5	2.32	5.97

Table B.3 Yearly Distribution of Legal Advisors per Deal

Year	<i>N</i>	Mean	Median	Std.	Min	P25	P75	Max	IQR	Range	Skew	Kurt.
1994	94	0.67	0.0	0.87	0	0.0	1.0	4	1.0	4	1.20	1.12
1995	192	0.83	0.0	1.01	0	0.0	2.0	4	2.0	4	0.96	0.09
1996	428	0.76	0.0	1.01	0	0.0	2.0	7	2.0	7	1.28	2.53
1997	516	0.82	0.0	0.98	0	0.0	2.0	5	2.0	5	0.86	-0.15
1998	677	0.83	0.0	0.97	0	0.0	2.0	4	2.0	4	0.77	-0.53
1999	660	0.95	1.0	0.99	0	0.0	2.0	5	2.0	5	0.67	-0.26
2000	685	1.01	1.0	1.01	0	0.0	2.0	5	2.0	5	0.66	-0.04
2001	538	0.98	1.0	1.05	0	0.0	2.0	7	2.0	7	1.07	2.16
2002	338	0.91	1.0	0.95	0	0.0	2.0	3	2.0	3	0.52	-1.01
2003	416	1.09	1.0	0.91	0	0.0	2.0	4	2.0	4	0.36	-0.62
2004	451	1.24	1.0	1.06	0	0.0	2.0	5	2.0	5	0.53	-0.05
2005	527	1.29	1.0	1.00	0	1.0	2.0	5	1.0	5	0.61	0.67
2006	688	1.31	1.0	0.98	0	1.0	2.0	5	1.0	5	0.59	0.28
2007	837	1.43	1.0	1.09	0	1.0	2.0	7	1.0	7	0.86	1.73
2008	517	1.59	2.0	1.13	0	1.0	2.0	7	1.0	7	0.70	1.90
2009	355	1.65	2.0	1.04	0	1.0	2.0	5	1.0	5	0.46	0.81
2010	485	1.75	2.0	1.14	0	1.0	2.0	6	1.0	6	0.68	0.91
2011	476	1.76	2.0	1.13	0	1.0	2.0	6	1.0	6	0.64	0.97
2012	400	1.74	2.0	0.94	0	1.0	2.0	7	1.0	7	0.35	1.86
2013	449	1.86	2.0	1.19	0	1.0	2.0	6	1.0	6	0.86	1.39
2014	491	1.98	2.0	1.24	0	1.0	2.5	9	1.5	9	0.97	3.01
2015	531	1.88	2.0	1.04	0	1.0	2.0	7	1.0	7	0.70	1.58
2016	545	1.98	2.0	1.15	0	1.0	3.0	9	2.0	9	0.94	3.15
2017	532	1.93	2.0	1.01	0	1.0	2.0	7	1.0	7	0.86	2.28
2018	515	2.05	2.0	1.13	0	1.0	2.0	8	1.0	8	1.02	2.48
2019	419	1.95	2.0	1.04	0	1.0	2.0	7	1.0	7	0.77	1.75
2020	355	2.03	2.0	1.18	0	1.0	2.0	7	1.0	7	1.03	2.12
2021	488	2.01	2.0	1.09	0	1.0	2.0	6	1.0	6	0.78	1.54
2022	382	1.86	2.0	0.93	0	1.0	2.0	5	1.0	5	0.34	0.57
2023	319	2.09	2.0	1.06	0	1.5	2.5	6	1.0	6	0.70	1.01
2024	367	1.99	2.0	1.07	0	1.0	2.0	7	1.0	7	0.95	1.61
2025	77	2.14	2.0	1.24	0	1.0	2.0	6	1.0	6	1.20	1.72